

Thesis Preparation

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These slides were in part prepared from the
materials of our colleague and friend
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slides are available at <https://magrilu.github.io/tht/>

Abstract

The abstract is a **small summary** of the thesis. It tells the reader in few words (up to one/one and a half page of total text) everything he/she needs to understand:

- the **context** of the work (e.g., chatbots),
- the **specific problem** approached by the thesis (e.g., the development of personal bots by non-programmers),
- if applicable, clearly state the research questions you would like to answer (e.g., "is it possible to enable non-programmers to do X using A?"),
- the **three/four core aspects** of the proposed solution (e.g., use predefined rules, use machine learning, assisted development, etc.),
- the **concrete outputs** produced by the thesis (e.g., a state of the art analysis, a conceptual/mathematical model, an application, middle-ware or API, an empirical study with/without users, etc.),
- the **findings and conclusions** that one can draw from the evaluation of the approach (e.g., that under some very specific conditions non-programmers are indeed able to implement own chatbots effectively using the proposed technique).

The Executive summary



- It's your calling card.
- It's the first thing that is read.
- You only get one chance to make a good first impression.

The Executive summary

The Executive Summary is a critical overview of your thesis with a focus on the main achievements that have emerged from your research.

In **6 pages** it has to describe:

- The **context** of your work
- The **problem** you are addressing and **why it is important**
- The **rationale** behind your solution
- The **Problem Formulation** (yes, equations!)
- The **key ingredients and a high-level description** of your solution (yes, equations and illustrations)
- The **most relevant findings** by presenting key experiments
- A short conclusion

Consider that your grade will be preliminary defined (or your reviewer will be primarily biased) from a skilled eye going through this document

The problem and its context

1. Introduction

Visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]).

The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance.

Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.



UNIVERSITÀ DI MILANO

Visual Localization in presence of match scarcity

Author: Francesco Basso
Advisor: Prof. Luca Sotgiu
Co-advisor: Prof. Roberto Cipriani
Co-advisor: Prof. Roberto Cipriani

1. Introduction

Visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]).

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1

Visual Localization

1. Problem Formulation

The problem of visual localization can be seen as a 3D reconstruction of a scene from a set of images. We assume the availability of a point cloud $\mathcal{P}$, a set of images $\mathcal{I}$, and a set of camera parameters $\mathcal{C}$. The goal is to find a camera pose $\mathbf{p}$ such that the image $\mathbf{p}$ is a perspective projection of the point cloud $\mathcal{P}$ through the camera parameters $\mathcal{C}$. This can be formulated as an optimization problem:

$$\min_{\mathbf{p}} \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{P}} \|\mathbf{p} - \mathbf{p}_i\|$$

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2



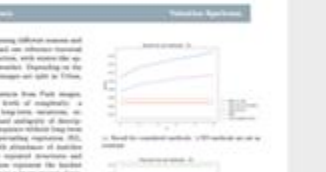
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5

Visual Localization



Figure 1: Visual Localization. The figure shows a query image being compared with a database of images to find matches. The process involves feature extraction, matching, and pose estimation.

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The goal is to find a camera pose $\mathbf{p}$ such that the image $\mathbf{p}$ is a perspective projection of the point cloud $\mathcal{P}$ through the camera parameters $\mathcal{C}$. This can be formulated as an optimization problem:

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3

Method	Rank 1 (%)	Rank 2 (%)	Rank 3 (%)	Rank 4 (%)	Rank 5 (%)
Baseline	10.0	15.0	20.0	25.0	30.0
Proposed	12.0	18.0	23.0	28.0	33.0

Table 1: Performance comparison on the test set.

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6

1. Introduction

Visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.

Definition of the problem

Visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driv-



POLITECNICO
MILANO (MI)

Visual localization for presence of safety security

Authors: Riccardo Basciani
Advisor: Prof. Luca Sordani
Co-advisor: Prof. Massimo Mucchetti, Dr. Giacomo Mucchetti
Students: Prof. Luca Sordani

1. Introduction
Visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.

2. Position Formulation
The problem of visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.

3. Related Work
The problem of visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.



Figure 1: A sequence of images showing a car driving through a tunnel, illustrating the concept of visual localization.

4. Proposed method
The problem of visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.



Figure 2: A diagram illustrating the process of visual localization, showing the relationship between the query image, the 3D reconstruction, and the estimated pose.

5. Experimental Results
The problem of visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.



Figure 3: A line graph showing the performance of different localization methods across various scenarios, with the proposed method showing superior results.

6. Conclusion
The problem of visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.

Method	Large-scale	Match scarcity
Proposed	92.5%	88.0%
Baseline 1	85.0%	80.0%
Baseline 2	80.0%	75.0%
Baseline 3	75.0%	70.0%

Table 1: Performance comparison across different scenarios.

7. Future Work
The problem of visual localization consists in inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.

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Why your problem is interesting? What is the impact of your research?

reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable.



POLITECNICO
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Visual localization to prevent road safety

Address: Department of Computer Science
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Address: Piazza S. Stefano

1. Introduction

Visual localization is the process of inferring the position and orientation from which a given picture (the *query*) was shot, by comparing it to a 3D reconstruction of a scene. As autonomous driving and augmented reality grow their popularity, more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the attained correspondences to solve for a pose. To prevent outliers from disrupting the estimate, a Perspective-n-Pose (PnP) solver is used within a robust fitting framework (e.g. RANSAC [2]). The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance. Yet, two settings remain critical to these methods: *large-scale* localization, where repetitive structures appear in different locations across the database, and localization with *match scarcity* in long-term scenarios, which entails few correct matches exist overall. In both cases performance drops, respectively by 7.4% and 44.3% of correct localizations.

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3. Related Work

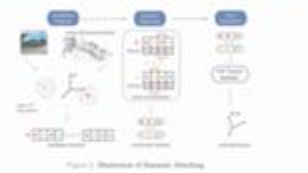
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Figure 1: A sequence of images showing a car driving on a road, illustrating the context of the problem.



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Method	Success Rate (%)	Execution Time (s)
Proposed	92.5	0.15
Baseline 1	85.0	0.20
Baseline 2	78.0	0.10

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What are the main challenges?

The problem becomes harder when radical appearance variations occur between reconstruction and query (*long-term variations*). State-of-the-art methods [3, 5] employ semantic filters to filter poor matches, because semantics are quite independent of the visual appearance.

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1. Introduction

Visual localization consists in inferring the position and orientation from which a given picture (the query) was shot, by comparing it to a 3D reconstruction of a scene. An autonomous driving and augmented reality gear that popularly more and more precise localization is desirable. Typically, visual localization is performed in two steps: (i) matching descriptors of the query and the 3D points of the scene, (ii) using the associated correspondence to solve for a pose. To prevent outliers from corrupting the estimate, a global optimization (e.g. RANSAC) is used. The problem becomes harder when reduced or poorer variations occur between consecutive images and long-term variations (e.g. RANSAC). One of the main methods [2, 3] employs semantic flows to filter poor matches, but semantic matches are quite sensitive to the visual appearance. We, therefore, consider in this work the use of the global information contained in the scene, rather than the absolute number of matches. The use of this method is in agreement with match scarcity in long-term variation, which entails few correct matches per scene. In fact, more than 50% of correct localizations.

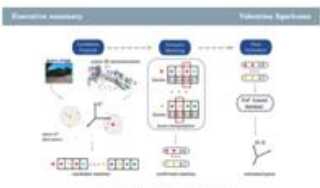


Figure 1: Illustration of Semantic Matching

of an individual match. The remaining uncertainty in the location of the camera center on a circle of radius r , with known orientation. This set of poses can be easily exploited for matching the position of the circle with known image in the scene. For each image, we can evaluate the number of matches M and the number of correct matches C . C and M are the number of matches C and M , and compare the score in Eq. 1.

We choose to evaluate the global semantic consistency as a percentage of correct matches in all projected points, differently from [5] who use the detection filter count. The choice is motivated by the fact that the number of projected points largely varies from match to match, making the absolute filter count hardly comparable for different matches.

The final choice of matches is performed by selecting the matches M with $C/M > \alpha$, where α is a threshold. The set of matches with $C/M > \alpha$ is then directly fed to RANSAC.

In Fig. 2, we provide a summary of the Semantic Matching procedure.

2.2. Global Semantic Matching for Robust Pose Estimation

Consider a collection of matches $M = \{m_1, \dots, m_n\}$ with associated semantic score C/M . The set of matches with $C/M > \alpha$ is then directly fed to RANSAC.

In Fig. 2, we provide a summary of the Semantic Matching procedure.

2.3. Distinct and Figure of merit

We use the Randomized Consensus (RANSAC) [6] for the set of long-term semantic matches. All images are taken in Pavia, Italy. The dataset consists of 100 images by 10

2. Problem Formulation

The input to visual localization algorithms are a query image and a 3D representation of the scene. We model the observation of a query image I_q as $I_q = I + N$, with I a set of feature descriptors $p_i = (x_i, y_i)$ ($x_i \in \mathbb{R}^2$, $y_i \in \mathbb{R}^2$). We also suppose to have a semantic match $S = \{1, \dots, L\}^{L \times L}$, with L a large number of classes $L \gg 10^3$, being given a list of global descriptors which is useful to verify global consistency to estimate match of semantic flows. A reconstruction of the scene, previously obtained from a collection of images $\{I_1, \dots, I_n\}$, is used to compute a pose (R, t) , with $R \in SO(3)$ representing the camera rotation and $t \in \mathbb{R}^3$ the translation vector of the camera center. The position of the pose is measured through the position p_i and the translation vector t (or p_i from the available ground truth pose (R_g, t_g)). These terms are defined as:

$$R_g = [C] - C_0, \quad C = -R^T t$$

$$t_g = [C] - C_0, \quad C = -R^T t$$

We assume we have for every image the global descriptor G as a camera coordinate and the camera height h . A challenge to long-term localization is represented by poor descriptor distinctiveness. The scene is a query descriptor G could have similar or even larger distance to the corresponding descriptor G_i of P_i than to the desired descriptor G_d of P_d . Moreover, pose estimation may be biased on the overall semantic consistency of the scene, rather than the absolute number of matches. The use of this method is in agreement with match scarcity in long-term variation, which entails few correct matches per scene. In fact, more than 50% of correct localizations.

3. Related Work

We review the most relevant researches on the use of semantic matching techniques and match consistency.

3.1. Matching in Visual Localization

Descriptors of appearance are commonly used to establish correspondences between images. These descriptors have desirable properties of robustness to scale and rotation changes, and distortions, that is a low probability of collision with unrelated descriptors thanks to their high dimensionality.

Matching and filtering are straightforward, since corresponding descriptors are expected to be close to each other, and for long-term localization, part of the last matches of long-term localization can be removed.

Subsequently, matches are filtered out by checking whether the ratio of distances to the first and second nearest neighbor is larger than some threshold (ratio test). The underlying principle of this approach is that most correspondences are within two categories: (i) correct matches, and (ii) incorrect matches which are far away from the query descriptor but are not corresponding among dataset points.

In long-term localization, the quality of the descriptor is more likely to vary due to increasing diversity in the set of descriptors. Long-term localization therefore requires these descriptors to be more robust to changes in appearance. For example, [8] include a robustness measure in the pose estimation, to account for pose variation. They use a robustness measure to filter out matches that are not robust to pose variation. They use a robustness measure to filter out matches that are not robust to pose variation. They use a robustness measure to filter out matches that are not robust to pose variation.



Figure 1: Example of a correct retrieval through Semantic Matching. The considered query image (bottom left) is associated to the wrong location due to the long-term variation of the described patch. The correct match is still close to the descriptor space and can be retrieved thanks to the semantic score.

various appearance pattern matching descriptors away from each other, and close to unrelated descriptors. Thanks to this observed property, part of the last matches of long-term localization can be removed.

3.2. Semantic Match Consistency

Semantic consistency is a suitable use of semantic in visual localization. It consists in checking of the labels of 3D points and their 3D correspondences are within two categories: (i) correct matches, and (ii) incorrect matches which are far away from the query descriptor but are not corresponding among dataset points.

Both the resulting RANSAC and GMM scores with the global information are used to filter matches for pose estimation. While this allows to improve the quality of the estimated pose, it also reduces the quality of the estimated pose. This is not possible to consider all the matches of every match in the pose estimation phase, but we should carefully select at most one match for all keypoints. Moreover, descriptor ambiguity can be a problem in long-term localization, where the scene is more likely to vary due to increasing diversity in the set of descriptors.

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In the executive summary you do not need too many references

1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications. It reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide-field-of-view (FV) cameras have become employed in those tasks, due to their increased angles of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images "in the wild".

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or at least

most part). However, there are situations where it is not possible to meet those constraints, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the Internet.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the downstream task. In this thesis, our contribution is two-fold. We first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a set of correspondences between the location of an image pixel and the light ray that reaches it, hence, it can represent any camera. Then, we devise a Deep Neural Network to estimate the proposed representation, replacing the radial

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of environments since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration and the robustness of our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from unknown to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera C in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\mathbf{K}_C$ of the camera C . The parameters $\mathbf{K}_C$ depend on the camera model M , while M must be specified as well. We focus on wide-angle cameras that have an angular field-of-view (AFV) greater than 90° .

Let us denote pixel coordinates as $\mathbf{u} = [u, v]^T \in \mathbb{C} \times \mathbb{C}$ and denote world 3D point coordinates as $\mathbf{x} = [x, y, z]^T \in \mathbb{C} \times \mathbb{C} \times \mathbb{C}$. The set $\mathcal{U}$ of the set of image points $\mathbf{u}$ that are hit by light rays (images have a finite number of pixels), and $\mathcal{U}$ is the set of 3D points $\mathbf{x}$ that are visible in the image (there may be world points that are not visible in the image). The relationship between a world point $\mathbf{x}$ and its corresponding image pixel $\mathbf{u}$ is determined by the geometric model of camera imaging, i.e., the Camera Model M . The camera model M defines a function $P_M: \mathcal{U} \rightarrow \mathbb{C}^3$, namely projection function, that depends on a set of parameters. The inverse is defined as $P_M^{-1}: \mathbb{C}^3 \rightarrow \mathcal{U}$ from image pixels to a bearing vector $\mathbf{u} \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world points $\mathbf{x}$ originating the

image point $\mathbf{u}$ cannot be recovered from a single image. The projection function P_M of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras, which are explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters $\mathbf{K}_C$, which model the optical properties of the camera, and the extrinsic parameters $\mathbf{C}_C$, that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\mathbf{K}_C = [f, c]^T$ of the input camera model M , where $f \in \mathbb{C}$ is the focal length and $c = [c_x, c_y]^T \in \mathbb{C}^2$ is the set of distortion parameters $\mathbf{K}_C$ and relative motion of the camera C . The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{x}_i)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

$$RE = \sum_{\mathbf{u} \in \mathcal{U}} \|\mathbf{u} - P_M(\mathbf{x}, \mathbf{K}_C)\|^2 \quad (1)$$

where $\mathcal{S}$ is the set of ground truth 3D-2D point correspondences $(\mathbf{x}, \mathbf{u})$, obtained knowing the ground truth projection function P_0 of the camera C .

$$S = \{\mathbf{x} \in \mathbb{C}^3, \mathbf{u} \in \mathcal{U} \mid P_0(\mathbf{x}) = \mathbf{u}\} \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Marker-based Calibration, Multi-view Autocalibration, and Monocular Autocalibration. In the following with the term camera we consider wide-angle cameras, since they are the focus of the research.

3.1. Marker-based Calibration
These are the most popular and widely used techniques calibration techniques. They are based on a known calibration target, typically a checkerboard [7], placed in the scene and observed under different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard corners) with a known position $\mathbf{x}$ in the scene, that are designed to be easy to detect in the image, in order

to recover their location $\mathbf{x}$. Thus, each marker provides a 3D-2D correspondence $(\mathbf{x}, \mathbf{u})$, that is used to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters $\mathbf{K}_C$. Our method severely improves the generalization of Marker-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model M from a sequence of images $I_0, I_1, \dots, I_n$. Those methods remove the need for a calibration target and jointly estimate the intrinsic parameters $\mathbf{K}_C$ and relative motion of the camera C . The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters $\mathbf{K}_C$ of an input camera model M . A branch of those methods, referred to as *geometric-driven*, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are straight lines [6], that allow imposing constraints on the vanishing points, from which the intrinsic parameters $\mathbf{K}_C$ can be derived. Those methods are suitable for man-made scenes, where such geometric cues are present in great numbers. The other branch of Monocular Autocalibration techniques leverages data-driven methods to learn complex features that are present in a wide range of scenes, without considering only straight lines of urban environments. The most well-known method is DeepCalib [2], which uses a CNN to directly regress the camera parameters $\mathbf{K}_C$ of a specific camera model M , the Unified Camera Model (UCM) [4], from the input image I . The DeepCalib [2] network is trained on a synthetically generated dataset, that includes

a broad range of scenes and camera configurations $\mathbf{K}_C, \mathbf{C}_C$. More recent methods have been proposed to partially estimate also the extrinsic parameters $\mathbf{C}_C$, predicting the tilt and roll of the camera C . We take inspiration from DeepCalib [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of cameras, using Polar coordinates, and that allows to calibrate of any camera model M since our network is trained to predict an intermediate camera representation.

4. Proposed Method

We present here the two main contributions of our work, the *Viewing-angle Intermediate Camera Representation (VICR)*, and the *Viewing-angle Calibration Method (VCM)*.

4.1. VICR

We adopt an implicit representation of camera that provides a mapping between image pixel $\mathbf{u}$ and world points $\mathbf{x}$. As opposed to camera models M , that explicitly model this mapping with complex mathematical functions, we rely on known 2D-3D correspondences $(\mathbf{u}_i, \mathbf{x}_i)$ between a subset of image pixels $\mathbf{u}$ and the bearing vector $\mathbf{x}$ of the light ray that hits such pixel $\mathbf{u}$. Thus, the VICR is able to represent any camera C and reduces approximation error introduced by mathematical camera models M . In practice, this intuitive definition of VICR is further simplified.

Without loss of generality, we consider an input image $I \in \mathbb{R}^{H \times W \times 3}$ with a square size, that can be obtained from a non-square image with the introduction of padding. Let $\mathbf{u} = [u_x, u_y] \in \mathbb{C}^2$ be an image pixel, we denote as $\mathbf{u}_0 = [u_x, u_y] \cdot \text{rad}$, the same image pixel in Polar coordinates, considering as pole the principal point $\mathbf{c} = [c_x, c_y]^T$. Then, let $\mathbf{x} = [x, y, z] \in \mathbb{C}^3$ be a world point, expressed in camera coordinates, since the extrinsic parameters $\mathbf{C}_C$ are known, we denote with $\mathbf{x}_0 = [x_0, y_0, z_0] \in \mathbb{C}^3$ the same point in spherical coordinates. As depicted in Figure 1, the center of the sphere coincides

with the origin of the camera frame, i.e., the camera center O . Given a generic correspondence $(\mathbf{u}, \mathbf{x})$ between an image pixel $\mathbf{u}$ and a world point $\mathbf{x}$, we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates $(\mathbf{u}, \mathbf{x})$, and remove the information regarding the distance of the point $\mathbf{x}$, obtaining $(\mathbf{u}, \mathbf{x}_0)$, where $\mathbf{x}_0 = (1, \sqrt{z})$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle ϕ of a light ray $\mathbf{x}_0$ is not altered by the camera. Thus, the camera optical distortions, the incidence angle $\sqrt{z}$ of the light ray $\mathbf{x}_0$, which modifies the value of $\mathbf{u}$ of the image point $\mathbf{u}_0$, and we rewrite the previous correspondence as $(\mathbf{u}, \sqrt{z})$. Let the viewing-angle $\mathcal{V}$ be a function that associates the value of a radial distance $\sqrt{z}$ to the incidence angle $\sqrt{z}$ of the light ray $\mathbf{x}_0$, defined as $\mathcal{V}: [0, \infty) \rightarrow [0, \pi/2]$, where ∞ is the maximum value of the radial distance $\sqrt{z}$, equal to half the image size $\sqrt{z} = S/2$. We can now define the VICR as a set of such correspondences $(\mathbf{u}, \mathcal{V})$, where the radial distance $\sqrt{z}$ is $\sqrt{z} = f(r, \theta)$. Each r_i is discrete, and R is uniformly distributed in the range $[0, \infty)$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C , and thus, we set $k = 25$.

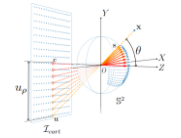


Figure 1: Illustration of the VICR.

4.2. VCM

We propose a method to calibrate the intrinsic parameters $\mathbf{K}_C$ from a single input image I , that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image I_{raw} . (ii) Both Cartesian I_{raw} and Polar I_{polar} images are given to a *Viewing-angle Estimation Network (VEN)* that predicts the VICR. The VEN is a CNN with two parallel feature extractors, namely CartFE and PolarFE, one for each input image. The extracted features are fused F_{fused} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors $\mathbf{K}_{\text{local}}$, one for each radial distance $r_i \in \mathbb{R}$ that compose the VICR. Each regressor $\mathbf{K}_{\text{local}}$ predicts the viewing-angle value $\mathcal{V}_i$ from a local slice of the extracted features F_{fused} , focusing on the image content near the radial distance r_i . (iii) As last, a *Viewing-angle Camera-model Fitting Algorithm* (VCFM) is employed to convert the VICR into a parameter configuration $\mathbf{K}_C$ for the input camera model M . Each correspondence $(\mathbf{u}, \mathcal{V})$ of the VICR can be seen as a sample of the radial projection function P_M of the camera model M , whose unknowns are the parameters $\mathbf{K}_C$. The VCFM obtains the parameters $\mathbf{K}_C$ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\mathcal{V}_i$ are strongly related to the content of the image at radius $\sqrt{z}$, compared to the target parameter $\mathbf{u}$ regression, where the targets are free scalar values that depend on the entire content of the image.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Recon-

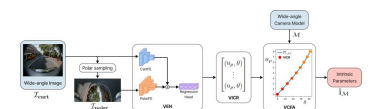


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample $(I, M, \mathbf{K}_C)$ 2-D of a calibrated dataset D , and from a randomly generated camera configuration $(M, \mathbf{K}_C)$, we use a mapping procedure to obtain a new image I_{new} and build the new dataset $(I_{\text{new}}, M, \mathbf{K}_C)$ 2-D. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [4] and SUN360 [8], two well-known Computer Vision dataset, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. KITTI-E2 is obtained from KITTI-360 [4], and contains outdoor scenes, both urban and landscape, thus with a limited presence of geometric cues. SUN-E2 is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. WSE-E2 contains urban scenes obtained from WoodScape [10], that provide a large viewpoint variation, finally, SLD-E2 also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_{sq} (Eq. 1) and the absolute reprojection error RE_{abs} (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences S is computed from the ground truth projection function $(M, \mathbf{K}_C)$, ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model M to be specified, while DeepCalib [2] supports only the UCM [8]. In these experiments, we used the Double Sphere camera model [8], since it is suitable for wide-ranged cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% over all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of Image Rectification (Figure 3). Starting from an input image I_{raw} (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\mathbf{K}_C$, and use them to produce the rectified image I_{rect} (Ours), where the distortion is no longer present. Comparing the rectified images I_{rect} are extremely similar to the ground truth image (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

Dataset	Method	RE _{sq} #		RE _{abs} #	
		Mean	Std	Mean	Std
KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35
	Ours	8.64	6.1	153.77	211.14
SUN-E2	DeepCalib	19.64	11.02	673.17	689.05
	Ours	13.79	9.4	402.07	517.97
SLD-E2	DeepCalib	14.74	10.24	413.49	521.75
	Ours	13.31	9.51	377.31	511.29
WSE2	DeepCalib	17.96	12.32	617.48	726.61
	Ours	15.94	9.54	495.42	511.99

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques, achieving a wide range of understanding in 2d and 3d. Pattern Analysis and Machine Intelligence (PAMI), 2022.

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most part). However, there are situations where it is not possible to meet these constraints, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the Internet.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the scene. In this thesis, we first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a set of correspondences between the location of an image pixel and the light ray that reaches it, hence, it can represent any camera. Then, we devise a Deep Neural Network to estimate the proposed representation, exploiting the radial

Executive summary

Francesco Azzone

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relationships between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model. The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of environments since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration results of our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from outdoor to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera C in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\mathbf{K}_C$ of the camera C . The parameters $\mathbf{K}_C$ depend on the camera model $\mathcal{M}$, which must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (AFV) greater than 80° . Let us denote pixel coordinates as $\mathbf{p} = [u, v]^T \in \mathbb{R} \times \mathbb{R}^2$ and denote world 3D point coordinates as $\mathbf{x} = [x, y, z]^T \in \mathbb{R} \times \mathbb{R}^3$. The subset Θ is the set of image points $\mathbf{p}$ that are hit by light rays (images have a finite number of pixels), and Ω is the set of 3D points $\mathbf{x}$ that result in valid projections (there may be world points that are not visible in the image). The relationship between a world point $\mathbf{x}$ and its corresponding image point $\mathbf{p}$ is determined by the geometric model of camera imaging, i.e., the Camera Model $\mathcal{M}$. The camera model $\mathcal{M}$ defines a function $P_{\mathcal{M}}: \Omega \rightarrow \Theta$, namely *projection function*, that depends on a set of parameters. The inverse is defined as $P_{\mathcal{M}}^{-1}: \Theta \rightarrow \Omega$ from image points $\mathbf{p}$ to bearing vectors $\mathbf{e} \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world points $\mathbf{x}$ originating the

image point $\mathbf{p}$ cannot be recovered from a single image. The projection function P of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras unless explicitly specified. Any camera model $\mathcal{M}$ has two sets of parameters: the *intrinsic* parameters $\mathbf{K}_C$, which model the optical properties of the camera, and the *extrinsic* parameters $\mathbf{e}_C$, that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\mathbf{K}_C = [f, d]$ of the input camera model $\mathcal{M}$, where $f \in \mathbb{R}$ is the focal length and $d = [d_1, \dots, d_4] \in \mathbb{R}^4$ is the set of distortion parameters of the model. We follow the common assumptions of zero skew ($\gamma = 0$), principal point placed in the sensor center ($\mathbf{c}_0 = W/2, c_y = H/2$), pixel aspect ratio equal to one ($f_x = f_y = f$), and disregard the extrinsic parameters $\mathbf{e}_C$. The goodness of the estimated intrinsic parameters $\hat{\mathbf{K}}_C$ is measured with the reprojection error RE , expressed as:

$$RE = \sum_{\mathbf{x} \in \Omega} \|\mathbf{u} - P_{\hat{\mathbf{K}}_C}(\mathbf{x}, \mathbf{u}_0)\|^2 \quad (1)$$

where S is the set of ground truth 3D-2D point correspondences $(\mathbf{x}, \mathbf{u})$ obtained, knowing the ground truth projection function $P_{\mathcal{M}}$ of the camera C .

$$\mathbf{u} = \mathbf{x} \odot \mathbf{e} \odot \mathbf{u}_0 \in \Theta \quad P_{\mathcal{M}}(\mathbf{x}, \mathbf{u}) = \mathbf{u} \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Markenshoef Calibration, Autocalibration, and Monocular Calibration. The latter can be further divided into two sub-areas: in the following with the term camera we consider wide-angle cameras, since the rest of the research is more general.

3.1. Markenshoef Calibration These are the most popular and widely used techniques calibration techniques. They are based on a known calibration target, typically a checkerboard [2], placed in the scene and observed under different viewpoints in several images. The calibration target contains several straight lines, in fact, the information about the known position $\mathbf{x}$ in the scene, that are designed to be easy to detected in the image, in order

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to recover their location $\mathbf{x}$. Thus, each marker provides a 2D-3D correspondence $(\mathbf{x}, \mathbf{u})$, that is used to constrain the projection function $P_{\mathcal{M}}$ of the input camera model $\mathcal{M}$, obtaining the intrinsic parameters $\mathbf{K}_C$. Our method severely improves the generalization of Markenshoef methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model $\mathcal{M}$ from a sequence of images $I_1, I_2, \dots, I_n$. These methods remove the need for a calibration target and jointly estimate the intrinsic parameters $\mathbf{K}_C$ and relative motion of the camera C . The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [4], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters $\mathbf{K}_C$ of an input camera model $\mathcal{M}$. A branch of these methods, named *geometric* methods, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are straight lines [8], that allow imposing constraints on the vanishing points, from which the intrinsic parameters can be recovered. Other methods are suitable for man-made scenes, where such geometric cues are present in great numbers. These methods leverage data-driven techniques to learn complex features that are present in a wide range of scenes, without considering only straight lines of urban environments. The most well-known method is DeepCalib [2], which uses a CNN to directly regress the camera parameters $\mathbf{K}_C$ of a specific camera model $\mathcal{M}$, the United Camera Model (UCM) [5], from the input image I . The DeepCalib method is trained on a synthetically generated dataset, that includes

a broad range of scenes and camera configurations $\mathbf{K}_C, \mathbf{e}_C$. More recent methods have been proposed to partially estimate also the extrinsic parameters $\mathbf{e}_C$, predicting the tilt and roll of the camera C . We take inspiration from DeepCalib [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of cameras, using Polar coordinates, and that allows to calibrate of any camera model $\mathcal{M}$ since our network is trained to predict an intermediate camera representation.

The use of Polar coordinates is inspired by [11], a learning-based method that is used to solve the image rectification problem. This problem is simpler than Monocular Autocalibration since it considers only the distortion (defined by parameters d) present in the input image I .

4. Proposed Method

We present here the two main contributions of our work, the *Viewing-angle Intermediate Camera Representation (VICR)*, and the *Viewing-angle Calibration Method (VCM)*.

4.1. VICR We adopt an implicit representation of cameras that provides a mapping between image point $\mathbf{p}$ and world points $\mathbf{x}$. As opposed to camera models $\mathcal{M}$, that explicitly model this mapping with complex mathematical functions, we only rely on known 2D-3D correspondences $(\mathbf{u}, \mathbf{x})$, between a subset of image pixels and the bearing vector $\mathbf{e}$ of the light ray that hits each pixel $\mathbf{p}$. Thus, the VICR is able to represent any camera C and reduces approximation error introduced by mathematical camera models $\mathcal{M}$. In practice, this intuitive definition of VICR is further simplified.

Without loss of generality, we consider an input image $I \in \mathbb{R}^{H \times W \times 3}$ with a square size, that can be obtained from a non-square image with the introduction of padding. Let $\mathbf{u} = [u, v]^T \in \Theta$ be an image pixel, we denote as $\mathbf{u}_0 = [u_0, v_0]^T \in \Theta$, the same image pixel in Polar coordinates, considering as pole the principal point $\mathbf{c} = [S/2, S/2]^T$. Then, let $\mathbf{x} = [x, y, z]^T \in \Omega$ a world point, expressed in camera coordinates, after the extrinsic parameters $\mathbf{e}_C$ are known, we denote with $\mathbf{x}_0 = [x_0, y_0, z_0]^T \in \Omega$, the same point in spherical coordinates. As depicted in Figure 1, a center of the sphere coincides

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with the origin of the camera frame, i.e., the camera center O . Given a generic correspondence $(\mathbf{u}, \mathbf{x})$ between an image point $\mathbf{u}$ and a world point $\mathbf{x}$, we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates $(\mathbf{u}, \mathbf{x})$, and remove the information regarding the distance of the point $\mathbf{x}$, obtaining $(\mathbf{u}, \mathbf{x}_0)$, where $\mathbf{x}_0 = (1, \sqrt{z})$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle ϕ of a light ray $\mathbf{s}$ is not altered after the radial distance of $\mathbf{u}$ of the image point $\mathbf{u}$, and we rewrite the previous correspondence as $(\mathbf{u}_0, \mathbf{x}_0)$. Let the viewing-angle β be a function that associates the value of a radial distance ρ to the incident angle $\phi = \beta(\rho)$, defined as $\beta(\rho) = \beta_0 + \beta_1 \rho$, where β_0, β_1 are the maximum value of the radial distance ρ , equal to half the image size $s = s/2$. We can now define the VICR as a set of such correspondences $(\mathbf{u}_0, \beta(\rho), \mathbf{x}_0)$, where the radial distances are $\rho \in \mathbb{R} \times \mathbb{R}^2$, $\mathbf{x}_0 = (1, \sqrt{z})$. Each ρ is discrete, and R is uniformly distributed in the range $[\rho_{\min}, \rho_{\max}]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C and thus, we set $k = 25$.

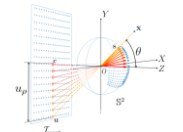


Figure 1: Illustration of the VICR.

4.2. VCM

We propose a method to calibrate the intrinsic parameters $\mathbf{K}_C$ from a single input image I , that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image I_{raw} . (ii) Both Cartesian I_{raw} and Polar I_{polar} images are given to a *Viewing-angle Estimation Network (VEN)* that predicts the VICR. The VEN is a CNN with two parallel branches: the Cartesian branch uses the PolarFE, one for each input image. The extracted features are fused F_{fused} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors M_{local} , one for each radial distance ρ . $\mathbb{R}^2$ that compose the VICR. Each regressor M_{local} predicts the viewing-angle value $\beta(\rho)$ from a local slice of the extracted features F_{fused} , focusing on the image content near the radial distance ρ . (iii) As last, a *Viewing-angle Camera-model Fitting Algorithm (VCFM)* is employed to convert the VICR into a parameter configuration $\mathbf{K}_C$ for the input camera model $\mathcal{M}$. Each correspondence $(\mathbf{u}_0, \beta(\rho), \mathbf{x}_0)$ can be seen as a sample of the radial projection function $P_{\mathcal{M}}$ of the camera model $\mathcal{M}$, whose unknowns are the parameters $\mathbf{K}_C$. The VCFM obtains the parameters $\mathbf{K}_C$ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\beta(\rho)$ are strongly related to the content of the image at radial distance ρ , compared to the parameter $\mathbf{u}_0$ region, where the targets are free scalar values that depend on the entire context of the images.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Rectification.

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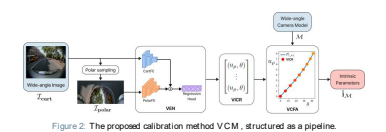


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample $(I, M, \mathbf{K}_C)$ 2-D of a calibrated dataset D , and from a randomly generated camera configuration $(M, \mathbf{K}_C)$, we use a mapping procedure to obtain a new image I_{new} , and build the new dataset $(I_{\text{new}}, M, \mathbf{K}_C)$ 2-D. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [4] and SUN360 [8], two well-known Computer Vision dataset, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. (KITTI-E2) is obtained from KITTI-360 [4], and contains outdoor scenes, both urban and landscapes, thus with a limited presence of geometric cues. (SUN-E2) is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. (WSE-E2) contains urban scenes obtained from WoodScape [10], that provide a large viewing-angle range. (SLD-E2) also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences S is computed from the ground truth projection $(M, \mathbf{K}_C)$, ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the camera model $\mathcal{M}$ to be specified, while DeepCalib [2] supports only the UCM [5]. In these experiments, we used the Double Sphere camera model [5], since it is suitable for wide-angle cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics on all the scenes. Note that the results of KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image I (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\mathbf{K}_C$, and use them to produce the rectified image I_r (Ours), where the distortion is no longer present. Comparing the images reported in Figure 3, we note our rectified images (Ours) are extremely similar to the ground truth ones (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

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Dataset	Method	RE _E #	RE _A #
KITTI-E2	Mean	20.18	11.47
	Std	711.22	706.35
	Ours	8.64	6.1
SUN-E2	Mean	11.02	6.73
	Std	673.17	689.05
	Ours	13.79	9.4
SLD-E2	Mean	14.74	10.24
	Std	413.49	521.75
	Ours	13.31	9.51
WSE-E2	Mean	17.96	12.32
	Std	617.48	726.61
	Ours	15.94	9.54

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the strong relationship with the wide-field-of-view cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics on all the scenes. Note that the results of KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

works.

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1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications, it reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide field-of-view (FOV) cameras have been employed in those tasks, due to their increased angles of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images “in the wild”.

Why is an important problem?

The problem and its context

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most part). However, there are situations where it is not possible to meet those requirements, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the Internet.

We solve the problem of calibrating a wide-angle camera using a simple, more reliable, and less challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the camera or the scene. In this thesis, our contribution is two-fold. We first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a set of correspondences between the location of an image pixel and the light ray that reaches it, hence, it can represent any camera. Then, we devise a Deep Neural Network to estimate the proposed representation, exploiting the radial

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or at least

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model. The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of environments since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration results of our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from outdoor to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera C in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\mathbf{K}_C$ of the camera C . The parameters $\mathbf{K}_C$ depend on the camera model M , which must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (AFov) greater than 80° . Let us denote pixel coordinates as $\mathbf{u} = [u, v]^T \in \mathbb{N} \times \mathbb{N}$ and denote world 3D point coordinates as $\mathbf{x} = [x, y, z]^T \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}$. The subset Θ is the set of image points $\mathbf{u}$ that are hit by light rays (images have a finite number of pixels), and Ω is the set of 3D points $\mathbf{x}$ that result in valid projections (there may be world points that are not visible in the image). The relationship between a world point $\mathbf{x}$ and its corresponding image pixel $\mathbf{u}$ is determined by the geometric model of camera imaging, i.e., the Camera Model M . The camera model M defines a function $P_M: \mathbb{R}^3 \rightarrow \Theta$, namely projection function, that depends on a set of parameters. The inverse is defined as $P_M^{-1}: \Theta \rightarrow \mathbb{R}^3$ from image pixels $\mathbf{u}$ to bearing vectors $\mathbf{x} \in \mathbb{R}^3$ of unit length, in fact, the information about the distance of the world point $\mathbf{x}$ originating the

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Our goal is to recover the intrinsic parameters $\mathbf{K}_C = [f, d]$ of the input camera model M , where $f \in \mathbb{R}$ is the focal length and $d = [d_1, \dots, d_4] \in \mathbb{R}^4$ is the set of distortion parameters of the model. We follow the common assumptions of zero skew ($\alpha = 0$), principal point placed in the sensor center ($c_x = W/2, c_y = H/2$), pixel aspect ratio equal to one ($f_x = f_y = f$), and disregard the extrinsic parameters $\mathbf{e}_C$. The goodness of the estimated intrinsic parameters $\hat{\mathbf{K}}_C$ is measured with the reprojection error RE, expressed as:

$$RE = \sum_{\mathbf{u} \in \Theta} \|\mathbf{u} - P_M(\hat{\mathbf{x}}, \hat{\mathbf{K}}_C)\|^2 \quad (1)$$

where $\hat{\mathbf{x}}$ is the set of ground truth 3D-point correspondences $(\mathbf{x}, \mathbf{u})$ obtained knowing the ground truth projection function P_M of the camera C .

$$\hat{\mathbf{x}} = [\mathbf{x} \oplus \mathbf{0}, \mathbf{u}] \in \Theta \times \Theta \times \mathbb{R}^3 \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Markenshaw Calibration, Autocalibration, and Monocular Calibration. Autocalibration is the process of recovering camera parameters from a sequence of images taken from different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard) with a known position $\mathbf{x}$ in the scene, that are designed to be easy to detect in the image, in order

to recover their location $\mathbf{x}$. Thus, each marker provides a 2D-3D correspondence $(\mathbf{x}, \mathbf{u})$, that is proposed to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters $\mathbf{K}_C$. Our method severely improves the generalization of Markenshaw methods, which is typically limited by the need for such calibration targets.

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We adopt an implicit representation of cameras that provides a mapping between image pixel $\mathbf{u}$ and world points $\mathbf{x}$. As opposed to camera models M , that explicitly model this mapping with complex mathematical functions, we only rely on known 2D-3D correspondences $(\mathbf{u}, \mathbf{x})$ between a subset of image pixels $\mathbf{u}$ and the bearing vector $\mathbf{x}$ of the light ray that hits such pixel $\mathbf{u}$. Thus, the VICR is able to represent any camera C and reduces approximation error introduced by mathematical camera models M . In practice, this intuitive definition of VICR is further simplified.

Without loss of generality, we consider an input image $I \in \mathbb{R}^{H \times W \times 3}$ with a square size, that can be obtained from a non-squared image with the introduction of padding. Let $\mathbf{u} = [u, v]^T \in \Theta$ be an image pixel, we denote as $\mathbf{u}_0 = [u_0, v_0]^T \in \Theta$, the same image pixel in Polar coordinates, considering as pole the principal point $\mathbf{c} = [c_x, c_y]^T \in \mathbb{R}^2$. Then, let $\mathbf{x} = [x, y, z]^T \in \mathbb{R}^3$ be a world point, expressed in camera coordinates, since the extrinsic parameters $\mathbf{e}_C$ are known, we denote with $\mathbf{x}_c = [\mathbf{x}_c, 0]^T \in \mathbb{R}^3$, the same point in spherical coordinates. As depicted in Figure 1, the center of the sphere coincides

with the origin of the camera frame, i.e., the camera center $\mathbf{O}$. Given a generic correspondence $(\mathbf{u}, \mathbf{x})$ between an image pixel $\mathbf{u}$ and a world point $\mathbf{x}$, we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates (u, x) , and remove the information regarding the distance of the point $\mathbf{x}$, obtaining (u, x) , where $x = (1, \angle \theta)$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle θ of a light ray $\mathbf{x}$ is not altered by the camera. Thus, the camera C only distorts the incidence angle $\sqrt{d}$ of the light ray $\mathbf{x}$, which modifies the distance of $\mathbf{u}$ of the image point $\mathbf{u}$, and we rewrite the previous correspondence as $(u, \sqrt{d})$. Let the viewing-angle $\sqrt{d}$ be a function that associates the value of a radial distance $\sqrt{d}$ to the incidence angle $\theta = \angle \theta$, defined as $\sqrt{d} = [0, \sqrt{d}]^T \in [0, \pi/2]$, where $\sqrt{d}_0$ is the maximum value of the radial distance $\sqrt{d}$, equal to half the image size $\sqrt{d}_0 = S/2$. We can now define the VICR as a set of such correspondences $(\mathbf{u}_0, \sqrt{d}_0)$, where the radial distances are $\sqrt{d} \in \mathbb{R} \times \mathbb{R}$, and the incidence angles $\theta \in [0, \pi/2]$. Each $\sqrt{d}_0$ is discrete, and $\mathbf{u}_0$ is uniformly distributed in the range $[0, \sqrt{d}_0]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C , and thus, we set $k = 25$.

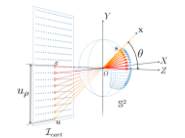


Figure 1: Illustration of the VICR.

4.2. VCM

We propose a method to calibrate the intrinsic parameters $\mathbf{K}_C$ from a single input image I , that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image I_{raw} . (ii) Both Cartesian I_{raw} and Polar I_{polar} images are given to a Viewing-angle Estimation Network (VEN) that predicts the VICR. The VEN is a CNN with two parallel branches: Cartesian Coordinates and PolarFE, one for each input image. The extracted features are fused F_{fused} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors M_{local} , one for each radial distance r , $r \in \mathbb{R}$ that compose the VICR. Each regressor M_{local} predicts the viewing-angle value $\sqrt{d}$ from a local slice of the extracted features F_{fused} , focusing on the image content near the radial distance r . (iii) As last, a Viewing-angle Camera-model Fitting Algorithm (VCFa) is employed to convert the VICR into a parameter configuration $\mathbf{K}_C$ for the input camera model M . Each correspondence $(\mathbf{u}_0, \sqrt{d}_0)$ can be seen as a sample of the radial projection function P_M of the camera model M , whose unknowns are the parameters $\mathbf{K}_C$. The VCFa obtains the parameters $\mathbf{K}_C$ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\sqrt{d}_0$ are strongly related to the content of the image at radius $\sqrt{d}_0$, compared to the radial parameter $\mathbf{u}_0$ regression, where the targets are free scalar values that depend on the entire content of the images.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Recon-

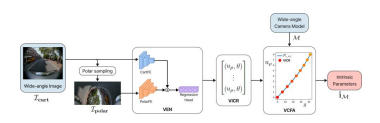


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample $(I, M, \mathbf{K}_C)$ 2 D of a calibrated dataset D , and from a randomly generated camera configuration $(M, \mathbf{K}_C)$, we use a mapping procedure to obtain a new image I_{new} , and build the new dataset $(I_{\text{new}}, M, \mathbf{K}_C)$ 2 G. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [4] and SUN360 [5], two well-known Computer Vision datasets, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. (KITTI-E2) is obtained from KITTI-360 [4], and contains outdoor scenes, both urban and landscapes, thus with a limited presence of geometric cues. (SUN-E2) is generated from a subset of SUN360 [5], and contains a wide range of indoor scenes. (WSE-E2) contains urban scenes obtained from WoodScape [10], that provide a large viewing-angle range, finally, (SLD-E2) also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences S is computed from the ground truth configuration $(M, \mathbf{K}_C)$, ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the camera model M to be specified, while DeepCalib [2] supports only the UCM [3]. In these experiments, we used the Double Sphere Camera Model [3], since it is suitable for wide-ranged cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image I_{raw} (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\mathbf{K}_C$, and use them to produce the rectified image I_{rect} (Ours), where the distortion is no longer present. Comparing the images reported in Figure 3, we note our rectified images (Ours) are extremely similar to the ground truth ones (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

Dataset	Method	RE _E #		RE _A #	
		Mean	Std	Mean	Std
KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35
	Ours	8.64	6.1	153.77	211.14
SUN-E2	DeepCalib	19.64	11.02	671.77	689.05
	Ours	13.79	9.4	402.07	517.87
SLD-E2	DeepCalib	14.74	10.24	413.49	521.75
	Ours	13.31	9.51	377.31	511.29
WSE2	DeepCalib	17.96	12.32	617.48	726.61
	Ours	15.94	9.54	498.42	511.99

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the wide-field-of-view cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

works.

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The problem and its context

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Executive Summary of the Thesis

Deep Monocular Autocalibration of Radially Symmetric Wide-Angle Cameras

Laura Magistrale in Computer Science and Engineering - Ingegneria Informatica

Author: Francesco Azzoni

Advisor: Prof. Luca Magri

Co-Advisor: Andrea Porri, Dal Cin, Prof. Giacomo Boracchi

Academic year: 2022-2023

1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications. It reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide-field-of-view (Fov) cameras have been employed in those tasks, due to their increased angle of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images "in the wild".

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or, at least, most part). However, there are situations where it is not possible to meet those constraints, e.g., in the case of images collected from cameras operated in environments subject to complex variations of temperature or humidity, or in the case of images collected from cameras operated in environments subject to complex variations of temperature or humidity.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the downstream task.

Our solution is a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNN with the radial projection function P_{polar} . (ii) As last, a View-angle Estimation Network (VEN) that predicts the VICR. The VEN is a CNN with two parallel branches: one for the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors M_{polar} , one for each radial distance r . $\mathbb{R}^2$ that compose the VICR. Each regressor M_{polar} predicts the viewing-angle value $\theta(r)$ from a local slice of the extracted features F_{polar} , focusing on the image content near the radial distance r . (iii) As last, a View-angle Camera-model Fitting Algorithm (VCF) is employed to convert the VICR into a parameter configuration μ for the input camera model M . Each correspondence (u, v) is transformed into a pair of polar coordinates (r, θ) . The VCF is composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C and thus, we set $k = 25$.

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with the origin of the camera frame, i.e., the camera center O . Given a generic correspondence (u, v) between an image point u and a world point v , we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates (u, v) , and remove the information regarding the distance of the point u , obtaining (u, v) , where $v = (1, \sqrt{v})$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle ϕ of a light ray s is not altered by the camera. Thus, the camera's only distortion is the incidence angle $\sqrt{v}$ of the light rays s , which modifies the radial distance of u of the image point u , and we rewrite the previous correspondence as $(u, \sqrt{v})$. Let the viewing-angle $\theta(\sqrt{v})$ be a function that associates the value of a radial distance $\sqrt{v}$ to the incidence angle $\theta(\sqrt{v})$, defined as $\theta(\sqrt{v}) = \theta(\sqrt{v})$, where $\theta(\sqrt{v})$ is the maximum value for the radial distance $\sqrt{v}$, equal to half the image size $\sqrt{v} = \sqrt{v}/2$. We can now define the VICR as a set of such correspondences $(u, \theta(\sqrt{v}))$, where the radial distances are $\sqrt{v} \in \mathbb{R}^2$, $\theta(\sqrt{v}) \in \mathbb{R}^2$. Each r_i is discrete, and R is uniformly distributed in the range $[0, \sqrt{v}]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C and thus, we set $k = 25$.

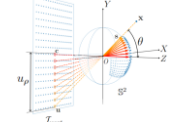


Figure 1: Illustration of the VICR.

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We propose a method to calibrate the intrinsic parameters μ from a single input image I , that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNN with the radial projection function P_{polar} . (ii) As last, a View-angle Estimation Network (VEN) that predicts the VICR. The VEN is a CNN with two parallel branches: one for the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors M_{polar} , one for each radial distance r . $\mathbb{R}^2$ that compose the VICR. Each regressor M_{polar} predicts the viewing-angle value $\theta(r)$ from a local slice of the extracted features F_{polar} , focusing on the image content near the radial distance r . (iii) As last, a View-angle Camera-model Fitting Algorithm (VCF) is employed to convert the VICR into a parameter configuration μ for the input camera model M . Each correspondence (u, v) is transformed into a pair of polar coordinates (r, θ) . The VCF is composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C and thus, we set $k = 25$.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Reconstruction.

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symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of scenarios since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration procedure of our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from unknown to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera C in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters μ of the camera C . The parameters μ depend on the camera model M , while μ must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (FOV) $\theta \in [0, \pi]$.

Let us denote pixel coordinates as $u = [u, v]^T \in \mathbb{C} \times \mathbb{R}^2$ and denote world 3D point coordinates as $x = [x, y, z]^T \in \mathbb{C} \times \mathbb{R}^3$. The set $\mathcal{U}$ is the set of image points u that are hit by light rays (images have a finite number of pixels), and $\mathcal{U}$ is the set of 3D points x that are visible in valid projections (there may be world points that are not visible in the image). The relationship between a world point x and its corresponding image pixel u is determined by the geometric model of camera imaging, i.e., the Camera Model M . The camera model M defines a function $P_M: \mathbb{C} \times \mathbb{R}^3 \rightarrow \mathbb{C} \times \mathbb{R}^2$, namely projection function, that depends on a set of parameters. The inverse is defined as $P_M^{-1}: \mathbb{C} \times \mathbb{R}^2 \rightarrow \mathbb{C} \times \mathbb{R}^3$ from image pixels to bearing vectors $s \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world points x originating the

image point u cannot be recovered from a single image. The projection function P_M of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras, which are explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters μ , which model the optical properties of the camera, and the extrinsic parameters e , that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\mu = [f, d]^T$ of the input camera model M , where $f \in \mathbb{R}$ is the focal length and $d \in [d_1, \dots, d_k] \in \mathbb{R}^k$ is the set of distortion parameters of the model. We follow the common assumption of zero skew ($\alpha = 0$), principal point placed in the sensor center ($c_x = W/2, c_y = H/2$), pixel aspect ratio equal to one ($f_x = f_y = f$), and disregard the extrinsic parameters e . The goodness of the estimated intrinsic parameters μ is measured with the reprojection error RE , expressed as:

$$RE = \sum_{u \in \mathcal{U}} \|u - P_M(x, \mu)\|^2 \quad (1)$$

where $\mathcal{S}$ is the set of ground truth 3D-2D point correspondences (x, u) , obtained knowing the ground truth projection function P_M of the camera C .

$$S = \{x \in \mathbb{C} \times \mathbb{R}^3, u \in \mathbb{C} \times \mathbb{R}^2 \mid x \sim u\} \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Marker-based Calibration, Multi-view Autocalibration, and Monocular Autocalibration. In the following with the term camera we consider wide-angle cameras, since they are the focus of the research.

3.1. Marker-based Calibration

These are the most popular and widely used techniques calibration techniques. They are based on a known calibration target, typically a checkerboard [7], placed in the scene and observed under different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard corners) with a known position x in the scene, that are designed to be easy to detect in the image, in order

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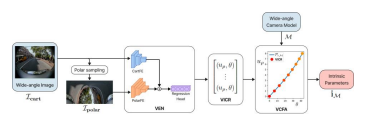


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample (I, M, μ, μ) 2 D of a calibrated dataset D , and from a randomly generated camera configuration (M, μ, μ) , we use a mapping procedure to obtain a new image I_{new} and build the new dataset $(I_{\text{new}}, M, \mu, \mu)$ 2 G. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [4] and SUN360 [8], two well-known Computer Vision dataset, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. (KITTI-E2) is obtained from KITTI-360 [4], and contains outdoor scenes, both urban and landscape, thus with a limited presence of geometric clues. (SUN-E2) is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. (WSE-E2) contains urban scenes obtained from WoodScape [10], that provide a large viewpoint variation, finally, (SLD-E2) also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences S is computed from the ground truth projection function (M, μ, μ) , ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model M to be specified, while DeepCalib [2] supports only the UCM [8]. In these experiments, we used the Double Sphere camera model [8], since it is suitable for wide-angle cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% over the baseline. On all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image I_{raw} (Original) that exhibits radial distortion, we employ our method to find the calibration parameters μ , and use them to produce the rectified image I_{rect} (Ours), where the distortion is no longer present. Comparing the image reported in Figure 3, we note our rectified images (Ours) are extremely similar to the ground truth ones (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

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to recover their location u . Thus, each marker provides a 2D-2D correspondence (x, u) , that is used to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters μ . Our method severely improves the generalization of Marker-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model M from a sequence of images $I_1, I_2, \dots, I_n$. These methods remove the need for a calibration target and jointly estimate the intrinsic parameters μ and relative motion of the camera C . The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [1], which uses point correspondences (u_i, u_j) between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters μ of an input camera model M . A branch of these methods, referred to as *geometric-driven*, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are straight lines [6], that allow imposing constraints on the vanishing points, from which the intrinsic parameters μ can be derived. These methods are suitable for man-made scenes, where such geometric clues are present in great numbers.

The other branch of Monocular Autocalibration techniques leverages data-driven techniques to learn complex features that are present in a wide range of scenes, without considering only straight lines of urban environments. The most well-known method is DeepCalib [2], which uses a CNN to directly regress the camera parameters μ of a specific camera model M , the Unified Camera Model (UCM) [4], from the input image I . The DeepCalib [2] network is trained on a synthetically generated dataset, that includes

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Dataset	Method	RE_#	Mean	Std	Mean	Std
KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35	
	Ours	8.64	6.1	153.77	211.14	
SUN-E2	DeepCalib	19.64	11.02	673.17	659.05	
	Ours	13.79	9.4	402.07	517.97	
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	Ours	15.94	9.54	496.42	511.99	

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the wide-field-of-view cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% over the baseline. On all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

6

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the downstream task.

Provide an idea of your solution

The problem and its context

1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications. It reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide-field-of-view (FOV) cameras have been employed in those tasks, due to their increased angles of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images "in the wild".

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or at least

most part). However, there are situations where it is not possible to meet those requirements, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the internet.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the scene.

We first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or at least

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symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of scenarios, since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration posed by our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from outdoor to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera C in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\mathbf{K}_C$ of the camera C . The parameters $\mathbf{K}_C$ depend on the camera model M , which must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (AFoV) greater than π .

Let us denote pixel coordinates as $\mathbf{u} = [u, v]^T \in \mathbb{C} \times \mathbb{R}$ and denote world 3D point coordinates as $\mathbf{x} = [x, y, z]^T \in \mathbb{C} \times \mathbb{R}^3$. The set $\mathcal{P}$ of the set of image points $\mathbf{u}$ that are hit by light rays (images have a finite number of pixels), and $\mathcal{L}$ is the set of 3D points $\mathbf{x}$ that are visible in valid projections (there may be world points that are not visible in the image).

The relationship between a world point $\mathbf{x}$ and its corresponding image pixel $\mathbf{u}$ is determined by the geometric model of camera imaging, i.e., the Camera Model M . The camera model thus defines a function $P_M: \mathcal{L} \rightarrow \mathcal{P}$, namely projection function, that depends on a set of parameters. The inverse is defined as $P_M^{-1}: \mathcal{P} \rightarrow \mathcal{L}$, from image pixels to a bearing vector $\mathbf{s} \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world point $\mathbf{x}$ originating the

image point $\mathbf{u}$ cannot be recovered from a single image. The projection function P_M of wide-angle cameras is often highly non-linear. We consider only camera models suitable for wide-angle cameras, which are explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters $\mathbf{K}_C$, which model the optical properties of the camera, and the extrinsic parameters $\mathbf{C}_C$ that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\mathbf{K}_C = [f, d]$ of the input camera model M , where $f \in \mathbb{R}$ is the focal length and $d = [d_1, \dots, d_6] \in \mathbb{R}^6$ is the set of distortion parameters of the model. We follow the common assumption of zero skew ($s = 0$), principal point placed in the sensor center ($c_0 = W/2, c_1 = H/2$), pixel aspect ratio equal to one ($d_5 = f_x = f_y$), and disregard the extrinsic parameters $\mathbf{C}_C$. The goodness of the estimated intrinsic parameters $\mathbf{K}_C$ is measured with the reprojection error RE , expressed as:

$$RE = \sum_{\mathbf{u} \in \mathcal{P}} \|\mathbf{u} - P_M(\mathbf{x}, \mathbf{K}_C)\|^2 \quad (1)$$

where $\mathcal{S}$ is the set of ground truth 3D-2D point correspondences $(\mathbf{x}, \mathbf{u})$, obtained knowing the ground truth projection function P_0 of the camera C .

$$S = \{(\mathbf{x}, \mathbf{u}) \in \mathcal{L} \times \mathcal{P} \mid P_0(\mathbf{x}) = \mathbf{u}\} \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Marker-based Calibration, Multi-view Autocalibration, and Monocular Autocalibration. In the following with the term camera we consider wide-angle cameras, since they are the focus of the research.

3.1. Marker-based Calibration

These are the most popular and widely used techniques calibration techniques. They are based on a known calibration target, typically a checkerboard [1], placed in the scene and observed under different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard corners) with a known position $\mathbf{x}$ in the scene, that are designed to be easy to detect in the image, in order

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to recover their location $\mathbf{x}$. Thus, each marker provides a 2D-2D correspondence $(\mathbf{x}, \mathbf{u})$, that is used to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters $\mathbf{K}_C$. Our method severely improves the generalization of Marker-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model M from a sequence of images $I_0, I_1, \dots, I_N$. These methods remove the need for a calibration target and jointly estimate the intrinsic parameters $\mathbf{K}_C$ and relative motion of the camera C . The methods proposed in the literature differ in the number of images N needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [2], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters $\mathbf{K}_C$ of the input camera model M . A branch of those methods, referred to as *geometric-driven*, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are straight lines [3], that allow imposing constraints on the vanishing points, from which the intrinsic parameters $\mathbf{K}_C$ can be derived. These methods are suitable for man-made scenes, where such geometric cues are present in great numbers.

The other branch of Monocular Autocalibration techniques leverages data-driven techniques to learn complex features that are present in a wide range of scenes, without considering only straight lines of urban environments. The most well-known method is DeepCalib [2], which uses a CNN to directly regress the camera parameters $\mathbf{K}_C$ of a specific camera model M , the Unified Camera Model (UCM) [4], from the input image I . The DeepCalib [2] network is trained on a synthetically generated dataset, that includes

a broad range of scenes and camera configurations $\mathbf{K}_C, \mathbf{C}_C$. More recent methods have been proposed to partially estimate also the extrinsic parameters $\mathbf{C}_C$, predicting the tilt and roll of the camera C . We take inspiration from DeepCalib [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of cameras, using Polar coordinates, and that allows to calibrate of any camera model M since our network is trained to predict an intermediate camera representation.

4. Proposed Method

We present here the two main contributions of our work: the *Viewing-angle Intermediate Camera Representation (VICR)*, and the *Viewing-angle Calibration Method (VCM)*.

4.1. VICR

We adopt an implicit representation of cameras that provides a mapping between image pixel $\mathbf{u}$ and world points $\mathbf{x}$. As opposed to camera models M , that explicitly model this mapping with complex mathematical functions, we rely only on known 2D-3D correspondences $(\mathbf{u}, \mathbf{x})$ between a subset of image pixels $\mathbf{u}$ and the bearing vector $\mathbf{s}$ of the light ray that hits each pixel $\mathbf{u}$. Thus, the VICR is able to represent any camera C and reduces approximation error introduced by mathematical camera models M . In practice, this intuitive definition of VICR is further simplified.

Without loss of generality, we consider an input image $I \in \mathbb{R}^{H \times W \times 3}$ with a square size, that can be obtained from a non-squared image with the introduction of padding. Let $\mathbf{u} = [u, v]^T \in \mathbb{C}$ be an image pixel, we denote as $\mathbf{u}_0 = [u_0, v_0]^T \in \mathbb{C}$ the same image pixel in Polar coordinates, considering as pole the principal point $\mathbf{c} = [c_0, c_1]^T$. Then, let $\mathbf{x} = [x, y, z]^T \in \mathbb{R}^3$ be a world point, expressed in camera coordinates, since the estimate parameters $\mathbf{e}$ are known, we denote with $\mathbf{x}_c = [x_c, y_c, z_c]^T \in \mathbb{R}^3$ the same point in spherical coordinates. As depicted in Figure 1, the center of the sphere coincides

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with the origin of the camera frame, i.e., the camera center O . Given a generic correspondence $(\mathbf{u}, \mathbf{x})$ between an image point $\mathbf{u}$ and a world point $\mathbf{x}$, we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates $(\mathbf{u}, \mathbf{x})$, and remove the information regarding the distance of the point $\mathbf{x}$, obtaining $(\mathbf{u}, \mathbf{s})$, where $\mathbf{s} = (1, \angle \mathbf{u})$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle ϕ of a light ray $\mathbf{s}$ is not altered by the camera. Thus, the camera C only distorts the incidence angle $\angle$ of the light ray $\mathbf{s}$, which modifies the radial distance $\|\mathbf{u}\|$ of the image point $\mathbf{u}$, and we rewrite the previous correspondence as $(\mathbf{u}, \angle \mathbf{u})$. Let the viewing-angle $\angle(\mathbf{u})$ be a function that associates the value of a radial distance $\|\mathbf{u}\|$ to the incidence angle $\angle \mathbf{u}$, defined as $\angle(\mathbf{u}) = \|\mathbf{u}\|^{-1} \cdot [0, 1]$, where $\|\mathbf{u}\|$ is the maximum value of the radial distance $\|\mathbf{u}\|$, equal to half the image size $\|\mathbf{u}\| = S/2$. We can now define the VICR as a set of such correspondences $(\mathbf{u}, \angle(\mathbf{u}))$, where the radial distances are $\|\mathbf{u}\| \in [0, S/2]$. Each r_i is discrete, and R is uniformly distributed in the range $[0, S/2]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C , and thus, we set $k = 25$.

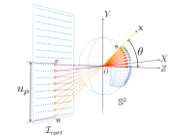


Figure 1: Illustration of the VICR.

4.2. VCM

We propose a method to calibrate the intrinsic parameters $\mathbf{K}_C$ from a single input image I , that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs with respect to the radial distribution of the distortion in the Cartesian image I_{raw} . (ii) Both Cartesian I_{raw} and Polar I_{polar} images are given to a *Viewing-angle Estimation Network (VEN)* that predicts the VICR. The VEN is a CNN with two parallel feature extractors, namely Cart2FE and Polar2FE, one for each input image. The extracted features are fused F_{fused} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors $\mathbf{K}_{\text{local}}$, one for each radial distance $r_i \in \mathbb{R}$ that compose the VICR. Each regressor $\mathbf{K}_{\text{local}}$ predicts the viewing-angle value $\angle(\mathbf{u})$ from a local slice of the extracted features F_{fused} , focusing on the image content near the radial distance r_i . (iii) As last, a *Viewing-angle Camera-model Fitting Algorithm* (VCF) is employed to convert the VICR into a parameter configuration $\mathbf{K}_C$ for the input camera model M . Each correspondence $(\mathbf{u}, \angle(\mathbf{u}))$ in the VICR can be seen as a sample of the radial projection function P_M of the camera model M , whose unknowns are the parameters $\mathbf{K}_C$. The VCF obtains the parameters $\mathbf{K}_C$ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\angle(\mathbf{u})$ are strongly related to the content of the image at radius $\|\mathbf{u}\|$, compared to the radial parameter $\|\mathbf{u}\|$ regression, where the targets are free scalar values that depend on the entire content of the image.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Recon-

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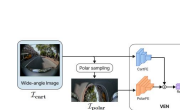


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample $(I, M, \mathbf{K}_C)$ 2-D of a calibrated dataset D , and from a randomly generated camera configuration $(M, \mathbf{K}_C)$, we use a mapping procedure to obtain a new image I_{new} and build the new dataset $(I_{\text{new}}, M, \mathbf{K}_C)$ 2-D. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [4] and SUN360 [8], two well-known Computer Vision dataset, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. (KITTI-E2) is obtained from KITTI-360 [4], and contains outdoor scenes, both urban and landscapes, thus with a limited presence of geometric cues. (SUN-E2) is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. (WSE-E2) contains urban scenes obtained from WoodScape [10], that provide a large viewpoint variation, finally, (SLD-E2) also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences S is computed from the ground truth configuration $(M, \mathbf{K}_C)$, ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model M to be specified, while DeepCalib [2] supports only the UCM [8]. In these experiments, we use the Double Sphere camera model [8], since it is suitable for wide-angle cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of Image Rectification (Figure 3). Starting from an input image I_0 (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\mathbf{K}_C$, and use them to produce the rectified image I_1 (Ours), where the distortion is no longer present. In comparison, the rectified image I_2 (DeepCalib) shows a significant residual distortion. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

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Dataset	Method	RE_E #	RE_A #
KITTI-E2	DeepCalib	20.18	1147
	Ours	8.64	6.1
SUN-E2	DeepCalib	19.64	1102
	Ours	13.79	9.4
SLD-E2	DeepCalib	14.74	1024
	Ours	13.31	9.51
WSE2	DeepCalib	17.96	1232
	Ours	15.94	9.54

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the wide-field-of-view cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

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1. Introduction

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Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same (or at least

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most part). However, there are situations where it is not possible to meet these constraints, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the Internet.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the environment task. In this thesis, our contribution is two-fold. We first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a set of correspondences between the location of an image pixel and the light ray that reaches it, hence, it can represent any camera. Then, we devise a Deep Neural Network to estimate the proposed representation, exploiting the radial

distortion as a pipeline, depicted in Figure 2. (i) The input image I , referenced to as I_{RAW} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{POLAR} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image I_{RAW} . (ii) Both Cartesian I_{RAW} and Polar I_{POLAR} images are given to a Viewing-angle Estimation Network (VEN) that predicts the VICR. The VEN is a CNN with two parallel branches: one for Cartesian and one for PolarFE, one for each input image. The extracted features are fused F_{FUSION} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors M_{LOCAL} , one for each radial distance $r_i \in \mathbb{R}^+$ that compose the VICR. Each regressor M_{LOCAL} predicts the viewing-angle value $\beta(r_i)$ from a local slice of the extracted features F_{FUSION} , focusing on the image content near the radial distance r_i . (iii) As last, a Viewing-angle Camera-model Fitting Algorithm (VCF) is employed to convert the VICR into a parameter configuration θ_{VCF} for the input camera model $\mathcal{M}$. Each correspondence $(u, v) \in \mathcal{C}$ can be seen as a pair of the radial projection function $P_{\mathcal{M}}$ of the camera model $\mathcal{M}$, whose unknowns are the parameters θ_{VCF} . The VCF is implemented by solving a simple non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\beta(r_i)$ are directly related to the content of the image at radius r_i , compared to the radial parameter r_i regression, where the targets are free scalar values that depend on the entire content of the image.

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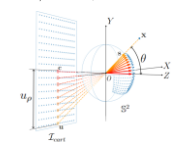


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The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of environments, since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration procedure of our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from unknown to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

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Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera $\mathcal{C}$ in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\theta_{\mathcal{M}}$ of the camera $\mathcal{C}$. The parameters $\theta_{\mathcal{M}}$ depend on the camera model $\mathcal{M}$, while must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (AFOV) greater than π . Let us denote pixel coordinates as $u = [u_x, u_y]^T \in \mathbb{C} \times \mathbb{C}$ and denote world 3D point coordinates as $x = [x_x, x_y, x_z]^T \in \mathbb{C} \times \mathbb{C} \times \mathbb{C}$. The set $\mathcal{C}$ is the set of image points u that are hit by light rays (images have a finite number of pixels), and Ω is the set of 3D points x that are visible in valid projections (there may be world points that are not visible in the image).

The relationship between a world point x and its corresponding image pixel u is determined by the geometric model of camera imaging, i.e., the Camera Model $\mathcal{M}$. The camera model $\mathcal{M}$ defines a function $P_{\mathcal{M}}: \Omega \rightarrow \mathbb{C}$, namely projection function, that depends on a set of parameters. The inverse is denoted as $P_{\mathcal{M}}^{-1}: \mathbb{C} \rightarrow \Omega$, from image pixels to bearing vectors $s \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world points x originating the

image point u cannot be recovered from a single image. The projection function $P_{\mathcal{M}}$ of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras, namely, we exclude specific. Any camera model $\mathcal{M}$ has two sets of parameters: the intrinsic parameters $\theta_{\mathcal{M}}$, which model the optical properties of the camera, and the extrinsic parameters $e_{\mathcal{M}}$ that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\theta_{\mathcal{M}} = [f, d]$ of the input camera model $\mathcal{M}$, where $f \in \mathbb{R}$ is the focal length and $d = [d_1, \dots, d_5] \in \mathbb{R}^5$ is the set of distortion parameters of the model. We follow the common assumption of zero skew ($s_x = 0$), principal point placed in the sensor center ($c_x = W/2, c_y = H/2$), pixel aspect ratio equal to one ($f_x = f_y = f$), and disregard the extrinsic parameters $e_{\mathcal{M}}$. The goodness of the estimated intrinsic parameters $\theta_{\mathcal{M}}$ is measured with the reprojection error RE, expressed as:

$$RE = \sum_{u \in \mathcal{C}} \|u - P_{\mathcal{M}}(x, \theta_{\mathcal{M}})\|^2 \quad (1)$$

where $\mathcal{S}$ is the set of ground truth 3D-2D point correspondences $(x, \theta_{\mathcal{M}})$ obtained knowing the ground truth projection function $P_{\mathcal{M}}$ of the camera $\mathcal{C}$.

$$S = \{x \in \Omega \mid u \in \mathcal{C}, P_{\mathcal{M}}(x) = u\} \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Marker-based Calibration, Multi-view Autocalibration, and Monocular Autocalibration. In the following with the term camera we consider wide-angle cameras, since they are the focus of the research.

3.1. Marker-based Calibration

These are the most popular and widely used techniques calibration techniques. They are based on a known calibration target, typically a checkerboard [1], placed in the scene and observed under different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard corners) with a known position in the scene, that are designed to be easy to detect in the image, in order

to recover their location u . Thus, each marker provides a 2D-3D correspondence (x, u) , that is used to constrain the projection function $P_{\mathcal{M}}$ of the input camera model $\mathcal{M}$, obtaining the intrinsic parameters $\theta_{\mathcal{M}}$. Our method severely improves the generalization of Marker-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model $\mathcal{M}$ from a sequence of images $I_0, I_1, \dots, I_N$. These methods remove the need for a calibration target and jointly estimate the intrinsic parameters $\theta_{\mathcal{M}}$ and relative motion of the camera $\mathcal{C}$. The methods proposed in the literature differ in the number of images N needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [3], which uses point correspondences (u_i, u_j) between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters $\theta_{\mathcal{M}}$ of an input camera model $\mathcal{M}$. A branch of those methods, referred to as *geometric-driven*, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are straight lines [4], that allow imposing constraints on the vanishing points, from which the intrinsic parameters $\theta_{\mathcal{M}}$ can be derived. These methods are suitable for man-made scenes, where such geometric cues are present in great numbers. The other branch of Monocular Autocalibration techniques leverages data-driven methods to learn complex features that are present in a wide range of scenes, without considering only straight lines of urban environments. The most well-known method is DeepCalib [2], which uses a CNN to directly regress the camera parameters $\theta_{\mathcal{M}}$ of a specific camera model $\mathcal{M}$, the United Camera Model (UCM) [5], from the input image I . The DeepCalib [2] network is trained on a synthetically generated dataset, that includes

without squaring the norm). The set of ground truth correspondences $\mathcal{S}$ is computed from the ground truth projection function $P_{\mathcal{M}}$, ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model $\mathcal{M}$ to be specified, while DeepCalib [2] supports only the UCM [5]. In these experiments, we use the Double Sphere camera model [6], since it is suitable for wide-angle cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% of F1 metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement is more significant on SLD-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image I_{RAW} (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\theta_{\mathcal{M}}$, and use them to produce the rectified image I_{RECT} (Ours), where the distortion is no longer present. Interestingly, SLD-E2 also contains distorted images, obtained from SLD-E2 [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the absolute reprojection error RE_A (Eq. 1) and the squared reprojection error RE_S (same as Eq. 1,

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6. Conclusions

In this work, we presented a novel camera representation that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the strong relationship with the wide-field-of-view cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% of F1 metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement is more significant on SLD-E2 and SLD-E2 is still remarkable and is completely unbiased.

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1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications. It reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide-field-of-view (FV) cameras have been employed in those tasks, due to their increased angles of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images "in the wild".

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or at least

most part). However, there are situations where it is not possible to meet those constraints, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the Internet.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, such estimates can be further refined with additional knowledge about the downstream task. In this thesis, our contribution is two-fold. We first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a set of correspondences between the location of an image pixel and the light ray that reaches it, hence, it can represent any camera. Then, we devise a Deep Neural Network to estimate the proposed representation, replacing the radial

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of scenarios, since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration results obtained by our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from unknown to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera $\mathbf{C}$ in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\mathbf{I}_M$ of the camera $\mathbf{C}$. The parameters $\mathbf{I}_M$ depend on the camera model $\mathbf{M}$, while $\mathbf{M}$ must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (AFV) greater than 90° .

Let us denote pixel coordinates as $\mathbf{u} = [u, v]^T \in \mathbb{C} \times \mathbb{C}$ and denote world 3D point coordinates as $\mathbf{x} = [x, y, z]^T \in \mathbb{C} \times \mathbb{C} \times \mathbb{C}$. The set $\mathbf{U}$ is the set of image points $\mathbf{u}$ that are hit by light rays (images have a finite number of pixels), and $\mathbf{U}$ is the set of 3D points $\mathbf{x}$ that are visible in the projections (there may be world points that are not visible in the image).

The relationship between a world point $\mathbf{x}$ and its corresponding image pixel $\mathbf{u}$ is determined by the geometric model of camera imaging, i.e., the Camera Model $\mathbf{M}$. The camera model $\mathbf{M}$ defines a function $P_M: \mathbb{C}^3 \rightarrow \mathbb{C}^2$, namely projection function, that depends on a set of parameters. The inverse is defined as $P_M^{-1}: \mathbb{C}^2 \rightarrow \mathbb{C}^3$ from image pixels to a bearing vector $\mathbf{s} \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world points $\mathbf{x}$ originating the

image point $\mathbf{u}$ cannot be recovered from a single image. The projection function P_M of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras, which are explicitly specified. Any camera model $\mathbf{M}$ has two sets of parameters: the intrinsic parameters $\mathbf{I}_M$, which model the optical properties of the camera, and the extrinsic parameters $\mathbf{E}_M$ that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\mathbf{I}_M = [f, d]$ of the input camera model $\mathbf{M}$, where f is the focal length and $d = [d_1, \dots, d_6] \in \mathbb{R}^6$ is the set of distortion parameters $\mathbf{I}_M$ and relative motion of the camera $\mathbf{C}$. The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{x}_i)$ between different images $(\mathbf{I}_i, \mathbf{I}_j)$ to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image $\mathbf{I}$.

$$RE = \sum_{\mathbf{u} \in \mathbf{U}} \|\mathbf{u} - P_M(\mathbf{x}, \mathbf{I}_M)\|^2 \quad (1)$$

where $\mathbf{S}$ is the set of ground truth 3D-2D point correspondences $(\mathbf{x}, \mathbf{u})$ obtained knowing the ground truth projection function P_M of the camera $\mathbf{C}$.

$$\mathbf{S} = \{(\mathbf{x}, \mathbf{u}) \in \mathbb{C}^3 \times \mathbb{C}^2 \mid P_M(\mathbf{x}) = \mathbf{u}\}$$

3. Related Works

We now explore the most relevant research area for our work, namely Marker-based Calibration, Multi-view Autocalibration, and Monocular Autocalibration. In the following with the term camera we consider wide-angle cameras, since they are the focus of the research.

3.1. Marker-based Calibration

These are the most popular and widely used techniques for camera calibration. They are based on a known calibration target, typically a checkerboard [7], placed in the scene and observed under different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard corners) with a known position $\mathbf{x}$ in the scene, that are designed to be easy to detect in the image, in order

to recover their location $\mathbf{x}$. Thus, each marker provides a 3D-2D correspondence $(\mathbf{x}, \mathbf{u})$, that is used to constrain the projection function P_M of the input camera model $\mathbf{M}$, obtaining the intrinsic parameters $\mathbf{I}_M$. Our method severely improves the generalization of Marker-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model $\mathbf{M}$ from a sequence of images $\mathbf{I}_0, \mathbf{I}_1, \dots, \mathbf{I}_n$. Those methods remove the need for a calibration target and jointly estimate the intrinsic parameters $\mathbf{I}_M$ and relative motion of the camera $\mathbf{C}$. The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images $(\mathbf{I}_i, \mathbf{I}_j)$ to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image $\mathbf{I}$.

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a broad range of scenes and camera configurations $\mathbf{C}_{CAM}$. More recent methods have been proposed to partially estimate also the extrinsic parameters $\mathbf{E}_M$, predicting the tilt and roll of the camera $\mathbf{C}$. We take inspiration from DeepCalib [2], and propose a novel solution that introduces new knowledge about the radial symmetry of scenes, using Polar coordinates, and that allows to calibrate of any camera model $\mathbf{M}$ since our network is trained to predict an intermediate scenes representation.

The use of Polar coordinates is inspired by [11], a learning-based method that is used to solve the image rectification problem. This problem is simpler than Monocular Autocalibration since it considers only the distortion (defined by parameters $\mathbf{d}$) present in the input image $\mathbf{I}$.

4. Proposed Method

We present here the two main contributions of our work, the *Viewing-angle Intermediate Camera Representation (VICR)*, and the *Viewing-angle Calibration Method (VCM)*.

4.1. VICR

We adopt an implicit representation of cameras that provides a mapping between image pixel $\mathbf{u}$ and world points $\mathbf{x}$. As opposed to camera models $\mathbf{M}$, that explicitly model this mapping with complex mathematical functions, we rely on known 2D-3D correspondences $(\mathbf{u}_i, \mathbf{x}_i)$ between a subset of image pixels $\mathbf{u}$ and the bearing vector $\mathbf{s}$ of the light ray that hits each pixel $\mathbf{u}$. Thus, the VICR is able to represent any camera $\mathbf{C}$ and reduces approximation error introduced by mathematical camera models $\mathbf{M}$. In practice, this intuitive definition of VICR is further simplified.

Without loss of generality, we consider an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ with a square size, that can be obtained from a non-squared image with the introduction of padding. Let $\mathbf{u} = [u, v]^T \in \mathbb{C} \times \mathbb{C}$ be an image pixel, we denote as $\mathbf{u}_i = [u_i, v_i]^T \in \mathbb{C} \times \mathbb{C}$, the same image pixel in Polar coordinates, considering as pole the principal point $\mathbf{c} = [c_x, c_y]^T \in \mathbb{C} \times \mathbb{C}$. Then, let $\mathbf{x} = [x, y, z]^T \in \mathbb{C} \times \mathbb{C} \times \mathbb{C}$ be a world point, expressed in camera coordinates, since the extrinsic parameters $\mathbf{E}_M$ are known, we denote with $\mathbf{x}_c = [x_c, y_c, z_c]^T \in \mathbb{C} \times \mathbb{C} \times \mathbb{C}$, the same point in spherical coordinates. As depicted in Figure 1, the center of the sphere coincides

with the origin of the camera frame, i.e., the camera center $\mathbf{O}$. Given a generic correspondence $(\mathbf{u}, \mathbf{x})$ between an image point $\mathbf{u}$ and a world point $\mathbf{x}$, we remove the information that is not related with the intrinsic properties of camera $\mathbf{C}$. We express the correspondence in polar and spherical coordinates $(\mathbf{u}, \mathbf{x})$, and remove the information regarding the distance of the point $\mathbf{x}$, obtaining $(\mathbf{u}, \mathbf{s})$, where $\mathbf{s} = (1, \angle \theta)$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera $\mathbf{C}$ exhibits a perfect radial symmetry, the azimuthal angle θ of a light ray $\mathbf{s}$ is not altered by the camera. Thus, the camera optical distortions, the incidence angle $\angle \theta$ of the light ray $\mathbf{s}$, which modifies the radial distance ρ of the image point $\mathbf{u}$, and we rewrite the previous correspondence as $(\mathbf{u}, \angle \theta)$. Let the viewing-angle $\angle \theta$ be a function that associates the value of a radial distance ρ to the incidence angle $\angle \theta$, $\angle \theta = \angle \theta(\rho)$, defined as $\angle \theta(\rho) = \arccos(\rho / \rho_0)$, where ρ_0 is the maximum value of the radial distance ρ , equal to half the image size $\rho_0 = S/2$. We can now define the VICR as a set of such correspondences $(\mathbf{u}, \angle \theta)$, where the radial distances are $\rho \in [0, \rho_0]$. Each ρ_i is discrete, and ρ is uniformly distributed in the range $[0, \rho_0]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera $\mathbf{C}$, and thus, we set $k = 25$.

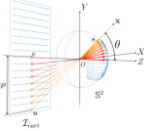


Figure 1: Illustration of the VICR.

4.2. VCM

We propose a method to calibrate the intrinsic parameters $\mathbf{I}_M$ from a single input image $\mathbf{I}$, that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image $\mathbf{I}$, referred to as $\mathbf{I}_{RAW}$, since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image $\mathbf{I}_{POL}$. In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image $\mathbf{I}_{RAW}$. (ii) Both Cartesian $\mathbf{I}_{RAW}$ and Polar $\mathbf{I}_{POL}$ images are given to a *Viewing-angle Estimation Network (VEN)* that predicts the VICR. The VEN is a CNN with two parallel feature extractors, namely CartFE and PolarFE, one for each input image. The extracted features are fused F_{FUSE} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors $\mathbf{I}_{M_{POL}}$, one for each radial distance $\rho_i \in \mathbb{R}$ that compose the VICR. Each regressor $\mathbf{I}_{M_{POL}}$ predicts the viewing-angle value $\angle \theta(\rho_i)$ from a local slice of the extracted features $F_{M_{POL}}$, focusing on the image content near the radial distance ρ_i . (iii) As last, a *Viewing-angle Camera-model Fitting Algorithm* (VCFM) is employed to convert the VICR into a parameter configuration $\mathbf{I}_M$ for the input camera model $\mathbf{M}$. Each correspondence $(\mathbf{u}, \angle \theta)$ of the VICR can be seen as a sample of the radial projection function P_M of the camera model $\mathbf{M}$, whose unknowns are the parameters $\mathbf{I}_M$. The VCFM obtains the parameters $\mathbf{I}_M$ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\angle \theta(\rho_i)$ are strongly related to the content of the image at radius ρ_i , compared to the target parameter $\mathbf{I}_M$ regression, where the targets are free scalar values that depend on the entire content of the image.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Recon-

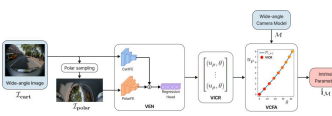


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample $(\mathbf{I}_i, \mathbf{M}_i, \mathbf{u}_i)$ 2 D of a calibrated dataset $\mathbf{D}$, and from a randomly generated camera configuration $(\mathbf{M}_j, \mathbf{u}_j)$, we use a mapping procedure to obtain a new image $\mathbf{I}_j$, and build the new dataset $(\mathbf{I}_j, \mathbf{M}_j, \mathbf{u}_j)$ 2 G. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [4] and SUN360 [8], two well-known Computer Vision datasets, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. (KITTI-E2) is obtained from KITTI-360 [4], and contains outdoor scenes, both urban and landscape, thus with a limited presence of geometric cues. (SUN-E2) is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. (WSE-E2) contains urban scenes obtained from WoodScape [10], that provide a large viewpoint variation, finally, (SLD-E2) also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences $\mathbf{S}$ is computed from the ground truth configuration $(\mathbf{M}_j, \mathbf{u}_j)$, ensuring that the same number of correspondences is visible in each image $\mathbf{I}_j$.

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model $\mathbf{M}$ to be specified, while DeepCalib [2] supports only the UCM [8]. In these experiments, we use the Double Sphere camera model [8], since it is suitable for wide-ranged cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance improvement achieved on WSE-E2 and SLD-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image $\mathbf{I}_0$ (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\mathbf{I}_M$, and use them to produce the rectified image $\mathbf{I}_1$ (Ours), where the distortion is no longer present. Comparing the rectified images $\mathbf{I}_0$ and $\mathbf{I}_1$, we note that our rectified images $\mathbf{I}_1$ are extremely similar to the ground truth ones (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

Dataset	Method	RE_E #		RE_A #	
		Mean	Std	Mean	Std
KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35
	Ours	8.64	6.1	153.77	211.14
SUN-E2	DeepCalib	19.64	11.02	673.17	689.05
	Ours	13.79	9.4	402.07	517.97
SLD-E2	DeepCalib	14.74	10.24	413.49	521.75
	Ours	13.31	9.51	377.31	511.29
WSE2	DeepCalib	17.96	12.32	617.48	726.61
	Ours	15.94	9.54	495.42	511.99

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques for learning-based calibration methods, due to the stronger relationship with the image content.

works.

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- [5] Christopher Mei and Patrick Rives. Single view geometry in perspective: Camera calibration from planar grids. In Proceedings 2007 IEEE International Conference on Robotics and Automation, pages 3945-3950, 2007.

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand,

image point u cannot be recovered from a single image. The projection function P of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras unless explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters I , which model the

to recover their location u . Thus, each marker provides a 2D-2D correspondence (x, u) , that is used to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters I_M . Our method severely improves the generalization of Marker-based methods, which is trivially limited by the need for

a broad range of scenes and camera configurations $k_{c,u}$. More recent methods have been proposed to partially estimate also the extrinsic parameters $e_{c,u}$, predicting the tilt and roll of the camera C . We take inspiration from DeepCall [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of

used to constrain the projection function $\mathcal{P}_M$ of the input camera model $\mathcal{M}$, obtaining the intrinsic parameters $\mathbf{i}_M$. Our method severely improves the generalization of Marker-based methods, which is trivially limited by the need for such calibration targets.

Most methods follow the work of [3], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images $(\mathcal{I}_i, \mathcal{I}_j)$ to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image $\mathcal{I}$.

Related work

For each stream of related work identify the main limitations and state how you have overcome them.

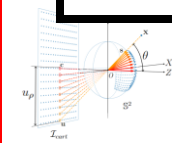


Figure 1: Illustration of the VICR.

problem. The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles (θ, r) are strongly related to the content of the image at radius u_r , compared to the direct parameter u_r regression, where the targets are few scalar values that depend on the entire content of the image.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Rectification.

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of environments since our solution is trained on a diverse set

image point u cannot be recovered from a single image. The projection function P of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras unless explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters i_M , which model the optical properties of the camera, and the extrinsic parameters e_M that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $i_M = [f, d]$ of the input camera model M , where $f \in \mathbb{R}$ is the focal length and $d = [d_1, \dots, d_4] \in \mathbb{R}^4$ is the set of distortion parameters of the

to recover their location u . Thus, each marker provides a 2D-2D correspondence (x, u) , that is used to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters i_M . Our method severely improves the generalization of Markers-based methods, which is trivially limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model M from a sequence of images $I_0, I_1, \dots, I_n$. Those methods remove the need for a calibration target and jointly estimate the intrinsic parameters i_M and

a broad range of scenes and camera configurations e_M . More recent methods have been proposed to partially estimate also the extrinsic parameters e_M , predicting the tilt and roll of the camera C . We take inspiration from DeepCalib [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of cameras, using Polar coordinates, and that allows to calibrate of any camera model M since our network is trained to predict an intermediate camera representation.

The use of Polar coordinates is inspired by [11], a learning-based method that is used to solve the image rectification problem. This problem is simpler than Monocular Autocalibration since

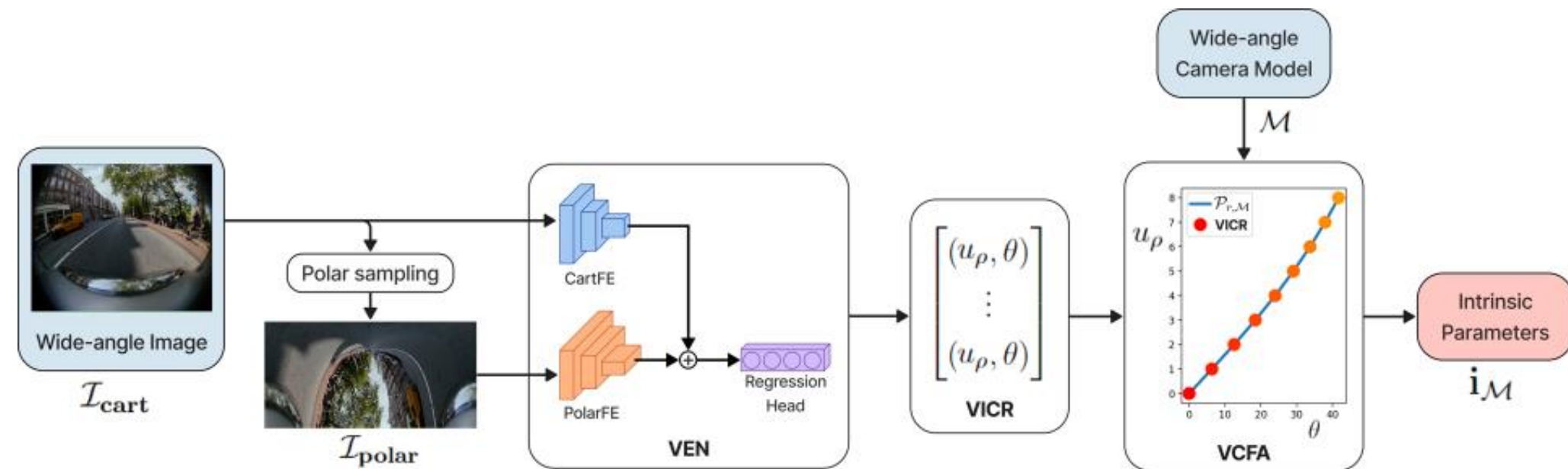


Figure 2: The proposed calibration method VCM, structured as a pipeline.

It's a good idea to have a nice illustration summarizing your method

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques on a wide range of conditions, for both the tasks of Camera Autocalibration and Image Rectification. Moreover, the CNN composing our solution is lighter than State-of-the-Art ones, achieving lower prediction times. Interesting future directions include exploiting the shape of objects in the scene to better understand the level of distortion introduced in the image, which could be done by leveraging segmentation net-

working mechanisms.

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[4] Yiwei Liao, Jun Xie, and Andreas Geiger. KITTI-360: A novel dataset and benchmarks for urban scene understanding in 2d and 3d. *Pattern Analysis and Machine Intelligence (PAMI)*, 2022.

[5] Christopher Mei and Patrick Rives. Single view point omnidirectional camera calibration from planar grids. In *Proceedings 2007 IEEE International Conference on Robotics and Automation*, pages 3945-3950, 2007.

1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications. It reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide-field-of-view (FOV) cameras have been employed in those tasks, due to their increased angles of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images "in the wild".

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or at least

most part). However, there are situations where it is not possible to meet those constraints, e.g., in the case of images collected from cameras operated in environments subject to consistent variations of temperature or humidity or collected from the Internet.

We solve the problem of calibrating a wide-angle camera using a single image, relaxing the constraints on the scene structure. This problem is challenging, since it includes a large variety of scenarios, and the available information is minimal. While the calibration results cannot be as precise as those obtained in controlled scenarios, still, an approximate estimation is enough to enable several applications. Moreover, our estimates can be further refined with additional knowledge about the downstream task. In this thesis, our contribution is two-fold. We first design a new camera representation that is directly related to the content of the image and avoids the use of any explicit mathematical model. Our representation is composed of a set of correspondences between the location of an image pixel and the light ray that reaches it, hence, it can represent any camera. Then, we devise a Deep Neural Network to estimate the proposed representation, exploiting the radial

distortion as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{raw} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{polar} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image I_{raw} . (ii) Both Cartesian I_{raw} and Polar I_{polar} images are given to the VEN Regression Head, which outputs the VEN, the VEN is a CNN with two parallel feature extractors, namely CartFE and PolarFE, one for each input image. The extracted features are fused F_{fused} in polar coordinates and given to the VEN Regression Head, which outputs the VEN. The Regression Head is composed of multiple local regressors M_{local} , one for each radial distance r , $r \in \mathbb{R}^+$ that compose the VICR. Each regressor M_{local} predicts the viewing-angle value $\beta(r)$ from a local slice of the extracted features F_{fused} , focusing on the image content near the radial distance r . (iii) As last, a Viewing-angle Camera-model Fitting Algorithm (VCF) is employed to convert the VICR into a parameter configuration θ for the input camera model M . Each correspondence (u, v) in the VICR is mapped to a point (u, v) in the radial projection function P_{θ} of the camera model M , whose unknowns are the parameters θ . The VCF obtains the parameters θ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\beta(r)$ are directly related to the content of the image at radius r , compared to the target parameter θ regression, where the targets are few scalar values that depend on the entire content of the images.

with the origin of the camera frame, i.e., the camera center O . Given a generic correspondence (u, v) between an image pixel u and a world point x , we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates (u, x) , and remove the information regarding the distance of the point x , obtaining (u, s_x) , where $s_x = (1, \sqrt{g})$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle ϕ of a light ray s_x is not altered by the camera. Thus, the camera-only distortion is the incidence angle $\sqrt{g}$ of the light ray s_x , which modifies the radial distance of u of the imaged point u , and we rewrite the previous correspondence as $(u, \sqrt{g})$. Let the viewing-angle $\beta(u)$ be a function that associates the value of a radial distance u , to the incidence angle $\sqrt{g}$, $\beta(u) = \sqrt{g}$, defined as $\beta(u) = \sqrt{g}$, $\beta(u) \in [0, \pi/2]$, where $\sqrt{g}$ is the maximum value for the radial distance u , equal to half the image size $u = S/2$. We can now define the VICR as a set of such correspondences $(u, \beta(u))$, where the radial distances are $u \in \mathbb{R}^+$ ($u = r_1, r_2, \dots, r_k$). Each r_i is discrete, and R is uniformly distributed in the range $[0, \sqrt{g}]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C , and thus, we set $k = 25$.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\beta(r)$ are directly related to the content of the image at radius r , compared to the target parameter θ regression, where the targets are few scalar values that depend on the entire content of the images.

5. Experiments
We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Rectification.

4.2. VCM
We propose a method to calibrate the intrinsic parameters θ from a single input image I , that leverages the VICR. Our method is struc-

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of real-world settings since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the estimates, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters θ of the camera C . The parameters θ depend on the camera model, which must be specified as input. We focus on wide-angle cameras that have an angular field-of-view (AFoV) greater than 90° .

Let us denote pixel coordinates as $u = [u_x, u_y]^T \in \mathbb{R}^2$, where S is the set of ground truth 3D-2D point correspondences $(\mathbf{x}, \mathbf{u})$, obtained knowing the ground truth projection function P_θ of the camera C .

image point u cannot be recovered from a single image. The projection function P_θ of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras unless explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters θ , which model the optical properties of the camera, and the extrinsic parameters e , that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters θ from a sequence of images $I_0, I_1, \dots, I_N$. Those methods remove the need for a calibration target and jointly estimate the intrinsic parameters θ and relative motion of the camera C . The methods proposed in the literature differ in the number of images N needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

$$RE = \sum_{\mathbf{u} \in S} \|\mathbf{u} - P_\theta(\mathbf{x}, \mathbf{u})\|^2 \quad (1)$$

where S is the set of ground truth 3D-2D point correspondences $(\mathbf{x}, \mathbf{u})$, obtained knowing the ground truth projection function P_θ of the camera C .

$$S = \{ \mathbf{x} \in \mathbb{R}^3, \mathbf{u} \in \mathbb{R}^2 | P_\theta(\mathbf{x}) = \mathbf{u} \} \quad (2)$$

3. Related Works

to recover their location $\mathbf{u}$. Thus, each method provides a 2D-2D correspondence $(\mathbf{x}, \mathbf{u})$, that is used to constrain the projection function P_θ of the input camera model M , obtaining the intrinsic parameters θ . Our method severely improves the generalization of Markov-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model M from a sequence of images $I_0, I_1, \dots, I_N$. Those methods remove the need for a calibration target and jointly estimate the intrinsic parameters θ and relative motion of the camera C . The methods proposed in the literature differ in the number of images N needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters θ of an input camera model M . A branch of those methods, referred to as *geometric-driven*, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are:

a broad range of scenes and camera configurations k_{CAM} . More recent methods have been proposed to partially estimate also the extrinsic parameters e_{CAM} , predicting the tilt and roll of the camera C . We take inspiration from DeepCalib [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of cameras, using Polar coordinates, and that allows to calibrate of any camera model M since our network is trained to predict an intermediate camera representation. The use of Polar coordinates is inspired by [11], a learning-based method that is used to solve the image rectification problem. This problem is simpler than Monocular Autocalibration since it considers only the distortion (defined by parameters d) present in the input image I .

4. Proposed Method

We present here the work of [8], which uses point correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

4.1. VICR

We adopt an implicit representation of camera that provides a mapping between image pixel u and world points x . As opposed to camera models M , that explicitly model this mapping with complex mathematical functions, we only rely on known 2D-3D correspondences $(\mathbf{u}_i, \mathbf{u}_j)$ between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

5.1. dataset and metrics

To evaluate the performance of tasks related to

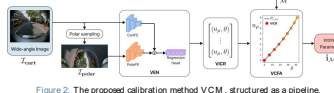


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample $(I, M, \theta, \mathbf{u})$ 2 D of a calibrated dataset D , and from a randomly generated camera configuration $(M, \theta, \mathbf{u})$, we use a mapping procedure to obtain a new image I_{new} and build the new dataset $(I_{\text{new}}, M, \theta, \mathbf{u})$ 2 G. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [6] and SUN360 [8], two well-known Computer Vision dataset, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. (KITTI-E2) is obtained from KITTI-360 [6], and contains outdoor scenes, both urban and landscapes, thus with a limited presence of geometric cues. (SUN-E2) is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. (WSE-E2) contains urban scenes obtained from WoodScape [10], that provide a large viewing-angle range. Finally, (SLD-E2) also contains urban images, obtained from SLDs [1], but it includes a broad range of weather and brightness conditions. We evaluate calibration results using the squared reprojection error RE_E (Eq. 1) and the absolute reprojection error RE_A (same as Eq. 1,

without squaring the norm). The set of ground truth correspondences S is computed from the ground truth projection function P_θ , ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model M to be specified, while DeepCalib [2] supports only the UCM [8]. In these experiments, we used the Double Sphere camera model [8], since it is suitable for wide-angle cameras. The results clearly confirm the validity of the proposed solution, reporting an improvement of at least 10% on all metrics, on all the scenes. Note that the results on KITTI-E2 and SUN-E2 are slightly biased in our favor, since they are generated from the same datasets used to generate the training set (of course, using different samples). On the other hand, the performance on WSE-E2 and SLD-E2 is more reliable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image I_{raw} (Original) that exhibits radial distortion, we employ our method to find the calibration parameters θ , and use them to produce the rectified image I_{rect} (Ours), where the distortion is no longer present. Comparing the images reported in Figure 3, we note our rectified images (Ours) are extremely similar to the ground truth ones (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35
	Ours	8.64	6.11	153.77	211.14
SUN-E2	DeepCalib	19.64	11.02	673.17	699.05
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Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.



Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques (WSE-E2) on all metrics, and is more robust to distortion, we employ our method to find the calibration parameters θ , and use them to produce the rectified image I_{rect} (Ours), where the distortion is no longer present. Comparing the images reported in Figure 3, we note our rectified images (Ours) are extremely similar to the ground truth ones (GT), significantly more than those obtained with DeepCalib [2]. This is a clear indication of the high quality of

Experiments

Present the dataset and the figure of merits used

works

References

- [1] Vasileios Balntas. SLDs: A Multi-Task Dataset for Evaluating Visual Localization. Medium, April 2019.
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1. Introduction

Camera calibration is a fundamental step for numerous Computer Vision applications. It reveals the relationship between image pixels and the location of the real-world object that they represent. This allows us to infer object sizes in world units, make distance measurements, and determine the location of a camera in the scene. Without this information, tasks such as 3D reconstruction, autonomous driving, visual localization, image rectification, and augmented reality would not be as successful. Recently, wide-field-of-view (Fov) cameras have been employed in those tasks, due to their increased angles of view. Their complex optical design introduces a severe distortion in the image, easy to note in presence of straight lines since they appear curved. Calibrating a wide-angle camera is a challenging task, especially considering images "in the wild".

Existing wide-angle calibration methods require a careful, controlled environment to be performed: they either need images of a calibration pattern (typically a checkerboard with accurate nominal measurements), assume the presence of straight lines in the scene, or require a sequence of images depicting the same scene (or, at least,

1

with the origin of the camera frame, i.e., the camera center O . Given a generic correspondence (u, x) between an image point u and a world point x , we remove the information that is not related with the intrinsic properties of camera C . We express the correspondence in polar and spherical coordinates (u, x) , and remove the information regarding the distance of the point x , obtaining (u, s) , where $s = (1, \sqrt{g})$ is a bearing vector, i.e., a vector with unitary norm (orange vectors in Figure 1). When camera C exhibits a perfect radial symmetry, the azimuthal angle ϕ of a light ray s is not altered by the camera. Thus, the camera's radial distortion is the incidence angle $\sqrt{d}$ of the light ray s , which modifies the radial distance r of the image point u , and we rewrite the previous correspondence as $(u, s) \rightarrow (u, \sqrt{d})$. Let the viewing-angle $\sqrt{d}$ be a function that associates the value of a radial distance r_u to the incidence angle $\sqrt{d} = \sqrt{d}(r_u)$, defined as $\sqrt{d}(r_u) = \sqrt{d}(r_u) \cdot \sqrt{d}(r_u)$, where $\sqrt{d}(r_u)$ is the maximum value for the radial distance r_u , equal to half the image size $u = u/2$. We can now define the VICR as a set of such correspondences $(u, \sqrt{d})$, where the radial distances are $u = 2 \cdot R \cdot (r_1, r_2, \dots, r_k)$. Each r_i is discrete, and R is uniformly distributed in the range $[0, \sqrt{d}(r_u)]$. We emphasized that a VICR composed by at least $k = 20$ correspondences is sufficient to precisely represent any camera C , and thus, we set $k = 25$.

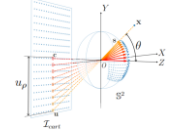


Figure 1: Illustration of the VICR.

4.2. VCM

We propose a method to calibrate the intrinsic parameters $\mathbf{I}_M$ from a single input image I , that leverages the VICR. Our method is struc-

tured as a pipeline, depicted in Figure 2. (i) The input image I , referred to as I_{RAW} , since it belongs to the Cartesian domain, is transformed into the Polar domain via the Polar sampling procedure, obtaining the Polar image I_{POLAR} . In the Polar domain, the distribution of the distortion depends only on the radial distance axis and is simpler to learn for CNNs compared to the radial distribution of the distortion in the Cartesian image I_{RAW} . (ii) Both Cartesian I_{RAW} and Polar I_{POLAR} images are given to a Viewing-angle Estimation Network (VEN) that predicts the VICR. The VEN is a CNN with two parallel feature extraction branches: CartNet and PolarFE, one for each input image. The extracted features are fused F_{FUSION} in polar coordinates and given to the VEN Regression Head, which outputs the VICR. The Regression Head is composed of multiple local regressors M_{LOCAL} , one for each radial distance r_i . R that compose the VICR. Each regressor M_{LOCAL} predicts the viewing-angle value $\sqrt{d}(r_i)$ from a local slice of the extracted features $F_{FEATURES}$, focusing on the image content near the radial distance r_i . (iii) As last, a Viewing-angle Camera-model Fitting Algorithm (VCF) is employed to convert the VICR into a parameter configuration $\mathbf{I}_M$ for the input camera model M . Each correspondence $(u, \sqrt{d})$ of the VICR can be seen as a sample of the radial projection function P_M of the camera model M , whose unknowns are the parameters $\mathbf{I}_M$. The VCF obtains the parameters $\mathbf{I}_M$ by simply solving a non-linear Least Squares fitting problem.

The proposed VICR plays a key role in the VCM. It facilitates the learning of the VEN since the targets viewing-angles $\sqrt{d}(r_i)$ are strongly related to the content of the image at radial distance r_i , compared to the polar parameter $\mathbf{I}_M$ regression, where the targets are fixed scalar values that depend on the entire content of the images.

5. Experiments

We conduct several experiments to assess the performance of the proposed VCM on the task of Monocular Autocalibration and Image Recon-

4

symmetry of wide-angle cameras, and a calibration algorithm to convert our representation into standard mathematical camera representations. On one hand, such an approach exploits data-driven techniques to learn complex relations between images and our representation, which cannot be modeled analytically. On the other hand, robust machine learning algorithms are used to filter out inaccurate estimates and to obtain a calibrated camera model.

The result is a calibration technique that can be employed for a wide range of cameras, thanks to the great expressiveness of our representation, and in a broad range of environments since our solution is trained on a large-scale dataset. We perform extensive experiments to assess the quality of the calibration pipeline of our method. Moreover, we compare its effectiveness against State-of-the-Art methods on diverse images, from outdoor to indoor, from urban to natural. The results show an improved calibration accuracy of over 10% compared to the best methods in the literature.

2. Problem Formulation

Given an image $I \in \mathbb{R}^{H \times W \times 3}$, taken by a wide-angle camera C in an uncontrolled environment, the goal of calibration is to compute the intrinsic parameters $\mathbf{I}_M$ of the camera C . The parameters $\mathbf{I}_M$ depend on the camera model M . $\mathbf{I}_M$ is the set of distortion parameters of the model. We focus on wide-angle cameras that have an angular field-of-view (AFov) greater than π .

Let us denote pixel coordinates as $u = [u, v]^T \in \mathbb{C} \times \mathbb{R}^2$ and denote world 3D point coordinates as $x = [x, y, z]^T \in \mathbb{C} \times \mathbb{R}^3$. The set $\mathcal{U}$ is the set of image points u that are hit by light rays (images have a finite number of pixels), and $\mathcal{U}$ is the set of 3D points x that are visible in valid projections (there may be world points that are not visible in the image).

The relationship between a world point x and its corresponding image point u is determined by the geometric model of camera imaging, i.e., the Camera Model M . The camera model M defines a function $P_M: \mathbb{C} \times \mathbb{R}^3 \rightarrow \mathbb{C} \times \mathbb{R}^2$, namely projection function, that depends on a set of parameters. The inverse is defined as $P_M^{-1}: \mathbb{C} \times \mathbb{R}^2 \rightarrow \mathbb{C} \times \mathbb{R}^3$ from image pixels to a bearing vector $s \in \mathbb{S}^2$ of unit length, in fact, the information about the distance of the world points x originating the

image point u cannot be recovered from a single image. The projection function P_M of wide-angle camera models is often highly non-linear. We consider only camera models suitable for wide-angle cameras, which are explicitly specified. Any camera model M has two sets of parameters: the intrinsic parameters $\mathbf{I}_M$, which model the optical properties of the camera, and the extrinsic parameters $\mathbf{E}_M$ that reveal the location and orientation of the camera in a world coordinate system.

Our goal is to recover the intrinsic parameters $\mathbf{I}_M = [f, d]$ of the input camera model M , where $f \in \mathbb{R}$ is the focal length and $d = [d_1, \dots, d_k] \in \mathbb{R}^k$ is the set of distortion parameters of the model. We make the common assumption of zero skew ($\alpha = 0$), principal point placed in the sensor center ($c_x = W/2, c_y = H/2$), pixel aspect ratio equal to one ($d_x = f_x = 1$), and disregard the extrinsic parameters $\mathbf{E}_M$. The goodness of the estimated intrinsic parameters $\mathbf{I}_M$ is measured with the reprojection error RE , expressed as:

$$RE = \sum_{u \in \mathcal{U}} \|u - P_M(x, \mathbf{I}_M)\|^2 \quad (1)$$

where $\mathcal{U}$ is the set of ground truth 3D-2D point correspondences (x, u) , obtained knowing the ground truth projection function P_M of the camera C .

$$S = \{x \in \mathbb{C} \times \mathbb{R}^3, u \in \mathbb{C} \times \mathbb{R}^2 \mid P_M(x) = u\} \quad (2)$$

3. Related Works

We now explore the most relevant research areas for our work, namely Marker-based Calibration, Multi-view Autocalibration, and Monocular Autocalibration. In the following with the term camera we consider wide-angle cameras, since they are the focus of the research.

3.1. Marker-based Calibration

These are the most popular and widely used techniques calibration techniques. They are based on a known calibration target, typically a checkerboard [7], placed in the scene and observed under different viewpoints in several images. The calibration target contains several markers (e.g., checkerboard corners) with a known position in the scene, that are designed to be easy to detect in the image, in order

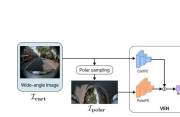


Figure 2: The proposed calibration method VCM, structured as a pipeline.

5.1. Dataset and metrics

To evaluate the performance of tasks related to camera calibration, we generated a large-scale dataset with a wide variety of scenes and cameras. This is a common choice in the literature, to overcome the lack of camera diversity in publicly available datasets, whose images are usually captured with the same camera. Starting from a sample (I, M, u) 2 D of a calibrated dataset D , and from a randomly generated camera configuration (M, u) , we use a mapping procedure to obtain a new image I_u , and build the new dataset (I_u, M, u) 2 D. We designed a custom configuration sampling strategy to reduce the number of ambiguous configurations, i.e., different configurations that model the same camera.

With this procedure, we obtained a large-scale dataset used to train the VEN, composed by 7,130K images generated from KITTI-360 [6] and SUN360 [8], two well-known Computer Vision datasets, that represents both outdoor and indoor, urban and natural scenes. Then, we generated four datasets for testing, each with a different type of scenario. KITTI-E2 is obtained from KITTI-360 [6], and contains outdoor scenes, both urban and landscape, thus with a limited presence of geometric cues. SUN-E2 is generated from a subset of SUN360 [8], and contains a wide range of indoor scenes. WS-E2 contains urban scenes obtained from WoodScape [10], that provide a large viewpoint variation, finally, SILDa-E2 also contains urban images, obtained from SILDa [1], but it includes a broad range of weather and brightness conditions.

We evaluate calibration results using the squared reprojection error RE_s (Eq. 1) and the absolute reprojection error RE_a (same as Eq. 1, without squaring the norm). The set of ground truth correspondences S is computed from the ground truth configuration (M, u) , ensuring that the same number of correspondences is visible in each image I .

5.2. Calibration experiments

We report the results of our method on the evaluation sets, compared with those of DeepCalib [2] (Table 1). Note that our method requires the input camera model M to be specified, while DeepCalib [2] supports only the UCM [8]. In these experiments, we used the Double Sphere camera model [8], since it is suitable for WS-E2 and SILDa-E2 is still remarkable and is completely unbiased.

5.3. Image Rectification Experiments

Our method is also compared with DeepCalib [2] on the task of image rectification (Figure 3). Starting from an input image I_u (Original) that exhibits radial distortion, we employ our method to find the calibration parameters $\mathbf{I}_M$, and use them to produce the rectified image I_r (Ours), where the distortion is no longer present. Comparing the results in Figure 3, we note that our rectified images (Ours) are extremely similar to the ground truth image (GT), significantly outperforming those obtained with DeepCalib [2]. This is a clear indication of the high quality of

to recover their location u . Thus, each marker provides a 2D-2D correspondence (x, u) , that is used to constrain the projection function P_M of the input camera model M , obtaining the intrinsic parameters $\mathbf{I}_M$. Our method severely improves the generalization of Marker-based methods, which is typically limited by the need for such calibration targets.

3.2. Multi-view Autocalibration

Several techniques have been proposed in the literature to calibrate a camera model M from a sequence of images $I_1, I_2, \dots, I_n$. These methods remove the need for a calibration target and jointly estimate the intrinsic parameters $\mathbf{I}_M$ and relative motion of the camera C . The methods proposed in the literature differ in the number of images n needed to find a solution and in the technique used to compute such solutions. Most methods follow the work of [3], which uses point correspondences (u_i, u_j) between different images (I_i, I_j) to formulate epipolar geometry constraints, retrieving the camera fundamental matrix. Our proposed solution improves the applicability of those methods since we require a single input image I .

3.3. Monocular Autocalibration

Monocular Autocalibration methods use a single input image I to estimate the intrinsic parameters $\mathbf{I}_M$ of an input camera model M . A branch of these methods, referred to as *geometric-driven*, exploits the presence of geometric structures in the scene to solve the problem. The most commonly used geometric elements are straight lines [6], that allow imposing constraints on the vanishing points, from which the intrinsic parameters $\mathbf{I}_M$ can be derived. These methods are suitable for man-made scenes, where such geometric cues are present in great numbers. The other branch of Monocular Autocalibration techniques leverages data-driven methods to learn complex features that are present in a wide range of scenes, without considering only straight lines of urban environments. The most well-known method is DeepCalib [2], which uses a CNN to directly regress the camera parameters $\mathbf{I}_M$ of a specific camera model M , the Unified Camera Model (UCM) [8], from the input image I . The DeepCalib [2] network is trained on a synthetically generated dataset, that includes

a broad range of scenes and camera configurations (x, u) . More recent methods have been proposed to partially estimate also the extrinsic parameters $\mathbf{E}_M$, predicting the tilt and roll of the camera C . We take inspiration from DeepCalib [2], and propose a novel solution that introduces prior knowledge about the radial symmetry of cameras, using Polar coordinates, and that allows to calibrate of any camera model M since our network is trained to predict an intermediate camera representation.

4. Proposed Method

We present here the two main contributions of our work: the *Viewing-angle Intermediate Camera Representation (VICR)*, and the *Viewing-angle Calibration Method (VCM)*.

4.1. VICR

We adopt an implicit representation of cameras that provides a mapping between image point u and world points x . As opposed to camera models M , that explicitly model this mapping with complex mathematical functions, we only rely on known 2D-2D correspondences (u_i, u_j) between a subset of image pixels u and the bearing vector s of the light ray that hits such pixel u . Thus, the VICR is able to represent any camera C and reduces approximation error introduced by mathematical camera models M . In practice, this intuitive definition of VICR is further simplified.

Without loss of generality, we consider an input image $I \in \mathbb{R}^{H \times W \times 3}$ with a square size, that can be obtained from a non-squared image with the introduction of padding. Let $u = [u, v]^T \in \mathbb{C} \times \mathbb{R}^2$ be an image pixel, we denote as $u = [u, v]^T \in \mathbb{C} \times \mathbb{R}^2$ the same image point in Polar coordinates, considering as pole the principal point $c = [c_x, c_y]^T \in \mathbb{C} \times \mathbb{R}^2$. Then, let $x = [x, y, z]^T \in \mathbb{C} \times \mathbb{R}^3$ be a world point, expressed in camera coordinates, since the extrinsic parameters $\mathbf{E}_M$ are known, we denote with $x_c = [x_c, y_c, z_c]^T \in \mathbb{C} \times \mathbb{R}^3$ the same point in spherical coordinates. As depicted in Figure 1, a center of the sphere coincides

Dataset	Method	RE _a #		RE _s #	
		Mean	Std	Mean	Std
KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35
	Ours	8.64	6.1	153.77	211.14
SUN-E2	DeepCalib	19.64	11.02	673.17	659.05
	Ours	13.79	9.4	402.07	517.97
SILDa-E2	DeepCalib	14.74	10.24	413.49	521.75
	Ours	13.31	9.51	377.31	511.29
WS-E2	DeepCalib	17.96	12.32	617.48	729.61
	Ours	15.94	9.54	496.42	511.99

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.

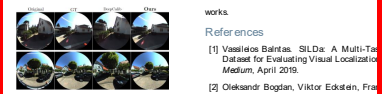


Figure 3: Image Rectification results on KITTI-E2.

our solution.

6. Conclusions

In this work, we presented a novel camera representation, that is more representative than the mathematical camera models, and is more suitable for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques for learning-based calibration methods, due to the stronger relationship with the image content. We also provide a new calibration method based on CNN, that leverages such representation, and make use of polar coordinates to simplify the problem. The proposed calibration method outperforms the State-of-the-Art techniques for learning-based calibration methods, due to the stronger relationship with the image content.

works

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- [4] Yi Li, Xian, Xie, and Andreas Geiger, KITTI-360: A novel dataset and benchmarks for a wide range of conditions in 3D, in *Pattern Analysis and Machine Intelligence (PAMI)*, 2022.
- [5] Christopher Mei and Patrick Rives, Single view pose estimation for camera calibration from planar grids, in *Proceedings 2007 IEEE International Conference on Robotics and Automation*, pages 3945–3950, 2007.

6

Experiments

Select some significant experiments that allow you to tell a story. Always draw conclusions!

Dataset	Method	RE _a #		RE _s #	
		Mean	Std	Mean	Std
KITTI-E2	DeepCalib	20.18	11.47	711.22	706.35
	Ours	8.64	6.1	153.77	211.14
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SILDa-E2	DeepCalib	14.74	10.24	413.49	521.75
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WS-E2	DeepCalib	17.96	12.32	617.48	729.61
	Ours	15.94	9.54	496.42	511.99

Table 1: Calibration results comparison with DeepCalib [2] on all of datasets.

The Introduction

- The introduction does not (anymore) corresponds to the executive summary!
- Describe the general context where your contribution takes place (e.g. smart devices for telemedicine... quality inspection...)
- Describe the goals
- Provide an outline of the thesis structure

The Introduction

<https://cs.stanford.edu/people/widom/paper-writing.html>

Here is the [Stanford InfoLab](#)'s patented five-point structure for Introductions. Unless there's a good argument against it, the Introduction should answer the following five questions:

1. *What is the problem?*
2. *Why is it interesting and important?*
3. *Why is it hard?* (E.g., why do naive approaches fail?)
4. *Why hasn't it been solved before?* (Or, what's wrong with previous proposed solutions? How does mine differ?)
5. *What are the key components of my approach and results?* Also include any specific limitations.

The Problem Formulation

The problem you are addressing need to be described in details. You have to clearly state

- What are the **inputs** you are given
 - If possible, provide an **observation model**
- The **output** (i.e. your goal)
- The **rule of the game**
 - **assumptions** you can leverage to reach the goal (e.g. **training data**)
 - **Specific contratints** (e.g. memory occupation, computing time...)

Related Work

Yes, everybody knows about perceptron... no need to go through that

Straight at the point:

- It should report the alternative approaches / solution to the problem you are addressing
- It should indicate the **originality** of your solution
- It should indicate **limitations** of other methods and how the proposed solution overcomes these
- It should pinpoint alternative methods
- It should mention also why other (related) solutions have not been considered

Related Work

Create a taxonomy

3. Related Works

3.1. Marker-based Calibration

3.2. Multi-view Autocalibration

3.3. Monocular Autocalibration

Peculiarities?
Benefits?

Limitations?

How my
method
overcome
the
limitations?



Typically, the benefits of your approach over alternatives must be assessed experimentally. Form the related work you get the baselines and the competitors of your method.

Technical Background

This is something you might need to set up the stage

You do **not need to make this apart from your contribution** (this is something you need to build up your work)

If short, it can be part of your «Proposed solution» section

Technical Background

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2.6	RANSAC	13
2.7	Epipolar geometry	16
3	Problem formulation	21

Proposed Solution



The right narrative: this is not a report of your work!

Clean up everything you have been doing, just illustrate your solution. -
All the failures / attempts / variants can be presented elsewhere

Take the **right technical level**: ideas and algorithms should be very neat and easy to grasp from a cursory look. Do not hide them in the mist of technical aspects. You can include an “Implementation Details” section

Do not expect the reviewer to study your thesis! It's up to you to make the few ideas and contribution to be very apparent.

Proposed Solution: Key Advices

You should:

- Use a few clear, nicely prepared and **commented illustrations**
- Use **Algorithms** that need to be commented in the text to illustrate your solution
- Use the **formal notation** of the problem formulation
- Recall the **problem formulation**
- Write with a **rigorous and technical style**

Proposed Solution

Always describe the algorithms in the text and reference their exact lines.

Make also reference to equations
`~\eqref{eq:replicas}`
`}`

In Algorithm 5 we iterate through each replicated vertex v in M (lines 2 - 14) and get the vector of replicas $\mathbf{m}_v$ for v (line 4). In lines 6 - 13 we encode the constraints for the replicas of v , as specified in Eq. (5.24). In line 8 we initialize $\mathbf{c}$ as a $dn \times 1$ vector of zeros and in lines 9 - 10 we update $\mathbf{c}$ in order to encode one of the equality constraints between the labels of the replicas $\mathbf{m}_v$ for v . The linear constraint in vertex form $\mathbf{c}$ is appended to $\mathbf{C}$ (line 11).

Algorithm 5 Building the matrix of linear constraints

Input: A mapping M between vertices in the underlying multigraph and their replicas in the expanded graph

Output: A matrix $\mathbf{C}$ containing the set of linear constraints that enforce the equality between the labels of replicas of the same vertex in the underlying multigraph

```
1:  $d$  = number of rows (columns) of matrices in  $GL(d)$ 
2: for every replicated vertex  $v \in M$  do
3:    $\triangleright$  Get the vector of vertices that are replicas of  $v$  in the expanded graph
4:    $\mathbf{m}_v = M[v]$ 
5:    $\triangleright$  Add linear constraints between the elements of replicas
6:   for  $j$  from 1 to  $d$  do
7:     for  $k$  from 1 to  $\text{length}(\mathbf{m}) - 1$  do
8:        $\mathbf{c} = \mathbf{0}_{dn}$   $\triangleright \mathbf{c}$  is a  $dn \times 1$  vector filled with zeros
9:        $\mathbf{c}(d \times \mathbf{m}_v[k] + j - d) = 1$ 
10:       $\mathbf{c}(d \times \mathbf{m}_v[k + 1] + j - d) = -1$ 
11:       $\mathbf{C} = [\mathbf{C}, \mathbf{c}]$ 
12:     end for
13:   end for
14: end for
```

Implementation Details

Not just a «collection of codes»

Nicely organized referring to the Proposed Solution, to «fill the missing gaps»

Adopt Algorithms, Illustrations, and a technical notation

Experiments

Typically there is no a single, simple, exhaustive, straightforward experiment.... You need to perform several experiments in specific setups

Overview of all the **experiments indicating their claim** (*e.g. we first verify that the proposed algorithm can preserve false alarms in ... and this is shown in Section 5.1. Then, ...*)

Experiments

Typically you might employ ad-hoc subsections for

- Datasets
- Figures of merit
- **Competing Methods** (and their configuration)
- One subsection per experiment

And, of course, include a specific Discussion subsection, or paragraphs

Experiments

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Conclusions

Summarize your major take home message

- What are your key contributions
- What are the key findings at the end of your experimental analysis

Illustrate future work

- We know that most probably you will be working @Google in the next few months....
- However, even if you won't continue working on this activity, you need to point out what you think is worth doing in the future, a some relevant contribution

Appendix

Technical details, additional experiments, general background can be move to the the appendix

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Writing Tips

Writing Style

Writing style

This is a M.Sc. thesis. It's neither Facebook nor Twitter nor an email. This is going to be an official document with legal value that will decide on the final mark of your yearlong university career and perhaps even on your future work perspectives. So, you surely don't want to be judged badly because of grammar errors, flawed/wrong vocabulary or superficial layout and/or text structure. It is a must that what you write is always correct content- and language-wise (no false statements or claims, no language mistakes), readable (no sentences that cannot be understood) and targeted at the *average-skilled reader* (professors, but also your own colleagues).

Reference to sections etc..

In Section~\ref{sec:intro}... Section when followed by a number, requires the capital letter.

*This applies also to Figure~\ref{fig:method},
Table~\ref{tab:exp1}, Chapter~\ref{ch:intro}...*

Plagiarism

Plagiarism

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Writing Tips - Paragraphs

Logically structured disposition (topic outline) is the most important thing in writing!

One paragraph, one key concept. Bear this in mind, it helps much in organizing your narrative. Readers expect that!

Writing Tips - Paragraphs

Each paragraph should have a purpose, should convey a message, should reflect a logical structure of your narrative.

... and should be nice!

Poor (you are not Joyce)

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Writing Tips - Punctuation

Do not use ; (or use them rarely... it's not C programming)

If any, use a single ; or :

If you prefer, you can write your thesis in Italian

- Use short sentences
- Use simple structures (direct form, use first person)
- Do not take this too personal.... You might use different words if something does not work.
- When some sentence simply does not work, just rewrite it from scratch, avoid minor, cosmetic changes
- Do not use synonyms for technical elements (do not talk about your solution/algorithm/approach/method interchangeably)
- Use formal notation
- Make notation simple, not cumbersome

$$x_{\alpha_i}^k$$

Writing Tips - Illustrations

Invest time in nice illustrations, which will be useful also for your presentation.

Illustrations should be well commented in the text. The rule of thumb is that the description should take as much space as the image itself.

Use captions.

If you use images done by others cite the source.

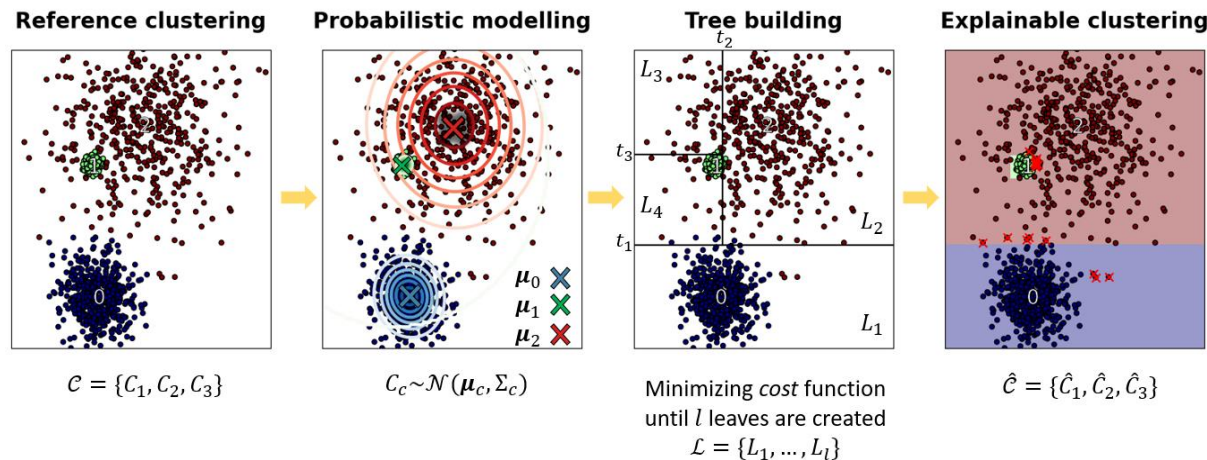


Figure 5.1: ELiCA in a nutshell. Given a reference clustering C as input, each cluster C_c is modeled as a Gaussian distribution, the likelihood of each cluster is hence used to build a binary threshold tree with at most l leaves, subdividing the space in rectangular regions that are used to induce an explainable clustering $\hat{C}$.

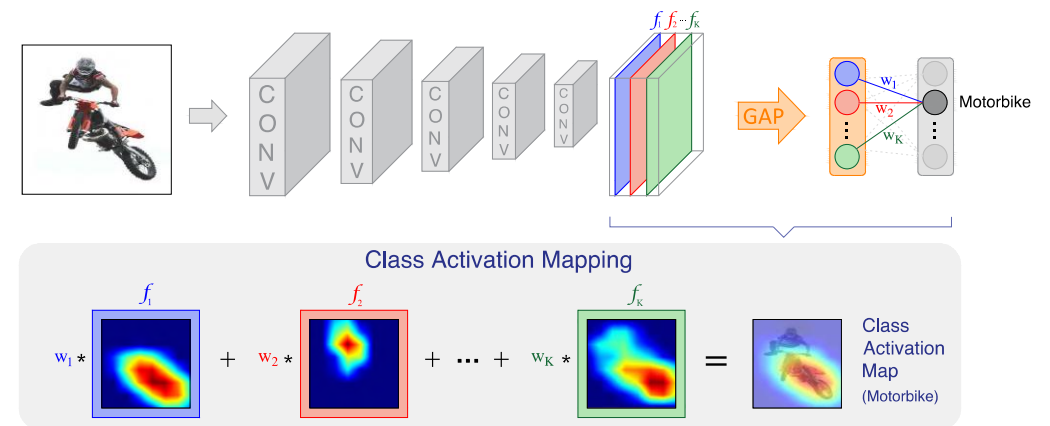


Fig. 4.17: In the upper part of the image the CNN architecture is shown. In the lower part the Class Activation Mapping technique is shown. Original figure from [32].

IOLO: I Only Look Once

... but answer as many questions as I can

Thesis revision is not an iterative process

vi beate i patica

